

Expecting Harm? The Impact of Rural Hospital Acquisitions on Maternal Health Care

David Dranove* Martin Gaynor†
Northwestern University Carnegie Mellon University and NBER
Eilidh Geddes‡
University of Georgia

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Abstract

While there is extensive research documenting the effects of hospital acquisitions on cost and quality, impacts on access to care have received less attention. Acquirers may limit access by closing an acquired hospital or eliminating individual service lines. However, acquirers may have more resources to improve care delivery. We study the impact of hospital acquisitions on obstetric care in rural markets, where there may be heightened concern about the availability of local care options. We find that acquired rural hospitals are substantially more likely to close obstetric units, with resulting large reductions in the number of births at the hospital. We find suggestive evidence of spillovers to general maternal health care resources and delayed prenatal care in the affected communities, but little evidence of effects on health outcomes. We find decreases in maternal transfers and increases in procedures consistent with women shifting to delivering in better resourced hospitals.

*d-dranove@kellogg.northwestern.edu

†mgaynor@cmu.edu

‡geddes@uga.edu

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1 Introduction

Experts describe rural health care in the United States as “in crisis” (e.g., [Bipartisan Policy Center, April, 2020](#); [Lopez, 2024](#)).¹ Rural hospital closures and service shutdowns contribute to the crisis ([U.S. Government Accountability Office, 2023](#)), forcing many patients to travel further or forego care altogether ([NIHCM Foundation, 2025](#)). At the same time, the number of physicians in rural areas has been declining ([U.S. Government Accountability Office; NIHCM Foundation](#)). The problem is exemplified by obstetric care. It is estimated that over two million women live in maternity care “deserts” ([Fontenot et al., 2024](#); [Kozhimannil, 2023](#)). These are counties, largely rural, that lack hospitals with obstetric units and professional obstetric providers. ([American Hospital Association, 2022](#); [Fontenot et al., 2024](#); [U.S. Government Accountability Office, 2022](#); [Kozhimannil et al., 2025](#)).

Reduced access may translate into poorer health outcomes, with growing disparities in life expectancy between urban and rural Americans ([Orozco Rodriguez, 2024](#)). This is again exemplified by obstetric care, as rural maternal and infant health outcomes are, in general, poorer than those in urban markets ([Nesbitt et al., 1990](#); [Kozhimannil et al., 2018](#); [Smith and Brown, 2024](#)).

Hospital acquisitions are an important part of this profound change in rural health care. Over the period 2007 to 2019, we identify 333 transactions where one or more rural hospitals are acquired. In total, 572 rural hospitals are acquired. Many of the targeted rural hospitals entirely close or eliminate important services, including obstetrics. In our sample, we observe 48 obstetric unit closures within 5 years of acquisition, a far higher closure rate than average for rural hospitals. Many of the shuttered units were the sole local source of obstetric care. After obstetric units close, pregnant women may face markedly higher travel costs. They may also receive less prenatal care, less timely obstetric care, more fragmented care, or

¹Also see <https://ruralhospitals.chqpr.org>

less postnatal care, any of which could result in worse outcomes for mothers or children ([Center for Health Care Quality and Payment Reform, 2023](#)). These trends raise concerns that acquisitions are contributing to the rural health crisis. In his 2021 Executive Order on Promoting Competition in the American Economy, President Biden noted how “hospital consolidation has left many areas, particularly rural communities, with inadequate or more expensive health care options.” ([Biden, 2021](#), Section 1).

Despite these concerns, closing obstetric units may yield some benefits. Closed units tend to be smaller and treat relatively few patients, and some studies document that complication rates and birth outcomes are poorer at low-volume hospitals ([Kyser et al., 2012](#); [Matsuo et al., 2022](#)). If acquirers redirect obstetrics patients to better quality hospitals, outcomes may improve.

Acquisitions of hospitals are part of a larger restructuring of the health economy, with over 1,300 mergers since 2000 ([Health Care Affordability Lab at Yale, 2026](#); [Gaynor, 2021](#)) and over 500 closures over a similar time period. As suggested by William Kissick’s seminal work “Medicine’s Dilemmas,” wholesale changes to the health care system, as embodied by the consolidation trend, can simultaneously affect costs, quality, and access to care ([Kissick, 1994](#)). Researchers have examined many of these effects, finding that consolidation generally leads to higher prices with ambiguous effects on quality. Most studies focus on urban markets and very few examine how consolidation affects access to care (see e.g., [Gaynor et al., 2015](#); [Handel and Ho, 2021](#)).² A handful examine the causes and consequences of rural hospital closures ([Hung et al., 2016](#); [Kozhimannil et al., 2018](#); [Jiang et al., 2022](#); [Carroll, 2023](#)), and the closures of specific units within rural hospitals, such as obstetrics ([Avdic et al., 2024](#); [Fischer et al., 2024](#); [Battaglia, 2025](#)). Again, only a few examine the resulting impact on access to care ([Henke et al., 2021](#); [O’Hanlon et al., 2019](#)).

Our primary interest is the impact of rural hospital acquisitions on access and outcomes.

²For a notable exception, see [Jiang et al. \(2021\)](#).

As a byproduct, we also compare the impact of acquisition-induced obstetric unit closures to research findings on the broader set of obstetric unit closures. Towards these ends, we assemble a panel of hospital mergers combining information about hospital service offerings, births, and staffing from the American Hospital Association’s (AHA) Annual Survey of Hospitals³ with the hospital merger database from Irving Levin Associates.⁴ We measure birth outcomes using county-level data from the National Vital Statistics System (NVSS).⁵

We estimate the impact of rural hospital acquisition using a difference-in-difference design, leveraging variation in the timing of acquisitions. We find that 5 years after acquisition, rural hospitals are 11 percent less likely to offer obstetric services than our control hospitals. Not surprisingly, there are large declines in the numbers of births at the acquired hospitals. We find suggestive evidence that closures following acquisition are more common when the transaction occurs within an HRR, the acquiring system contains no urban hospitals, and when the target hospital is located outside of a micropolitan area.

We find evidence of spillovers to other healthcare resources: we find decreases in obstetric care resources in the county in which the acquired hospital is located. This decline in local resources results in increased travel time for women. We estimate declines in initiation of prenatal care in the first trimester and in the recommended number of visits.

However, there is also evidence of increased use of anesthesia and decreases in maternal transfers, consistent with women delivering at better resourced hospitals. We find no evidence of changes in infant mortality or maternal morbidity or mortality. This contrasts with the literature on obstetric unit closures generally (Battaglia, 2025; Fischer et al., 2024) that finds improvements in maternal morbidity, although our estimates may reflect other changes from the acquisition rather than just the direct effects of the closure.

³<https://www.ahadata.com/aha-annual-survey-database>

⁴<https://www.levinassociates.com/health-care-services-acquisition-report/>

⁵National Center for Health Statistics, “Birth Files 2007-2019,” as compiled from data provided by the 57 vital statistics jurisdictions through the Vital Statistics Cooperative Program

Lastly, we compare acquisition-induced closures with the broader set of obstetric unit closures. We find that acquired hospitals tend to shutter larger obstetric units and are closer to alternative obstetric units. Overall, closures following acquisitions tend to have more muted effects on patient outcomes than the broader set of closures that have plagued rural healthcare, where patient outcomes have improved in some ways and suffered in others.

The rest of the paper is organized as follows. Section 2 provides further detail on the data that we use and presents some descriptive statistics. Section 3 describes our event study approach to estimating the effects of hospital acquisition on obstetric care. We then present our estimates of the impacts of rural hospital acquisitions on several outcomes in Section 4, taking account of possible heterogeneity and assessing the robustness of the results. Finally, we conclude in Section 5.

2 Data Sources and Descriptive Statistics

We use data from several sources that we describe in this section. Depending on the question at hand, we perform some analyses at the hospital level and others at the county level.

2.1 Hospital Data

Our hospital data primarily come from the American Hospital Association (AHA) Annual Survey of Hospitals from 2007-2019. Our sample consists of rural hospitals that are classified as either general hospitals or obstetrics and gynecology hospitals. We define rural hospitals as those that are located outside of Metropolitan Statistical Areas as defined by the Office of Management and Budget, following the Federal Office of Rural Health Policy’s definition of “rural”. The AHA data contain information on hospital characteristics, service lines, number of admissions, number of births, and staffing. We combine these data with hospital financial characteristics from the Healthcare Provider Cost Reporting Information System (HCRIS).

Obstetric Care – To identify hospitals that provide obstetric care, we use an algorithm similar to those described in [Interrante et al. \(2022\)](#) and [Fischer et al. \(2024\)](#) that use multiple variables from the AHA data to define whether a hospital provides obstetric care. We additionally validate this measure by constructing a measure of whether obstetric care is available at the county level using information on the location of births from the NVSS in a similar way to [Fischer et al. \(2024\)](#). We provide additional details on this process in Appendix [A.1](#).

Acquisitions – To identify all mergers and acquisitions, we detect all system changes in the AHA data and flag these as potential mergers. These include both the addition of new hospitals to existing systems and the creation of new systems. A hospital is defined as a target hospital (or an acquired hospital) if it is part of a merger where it is the smaller party and continues operating as a distinct facility for at least some time after the merger. We then link these mergers with Irving Levin (IL) Associates’ Healthcare M&A Report to verify the dates of system changes and to assist in identifying targets and acquirers. Additional details are available in Appendix [A.2](#).

2.2 County Level Outcomes Data

To measure patient outcomes, we use birth and mortality records from the National Vital Statistics System (NVSS). The NVSS data do not identify the birth hospitals, so we instead aggregate NVSS data and perform analyses at the county level. Our sample consists of rural counties from 2007-2019. We classify counties as rural if they are outside of Metropolitan Statistical Areas. For each outcome, we construct a balanced panel to remove counties where reporting changes over our sample period.

We measure the following: whether the mother delivers in her county of residence, c-section, induction, anesthesia, and maternal transfer rates, infant and maternal mortality, and indices of maternal morbidity. We aggregate conditions together into these indices as

each condition alone is relatively rare. These indices include a measure of severe maternal morbidity as reported on the birth certificate (maternal transfusion, perineal laceration, ruptured uterus, unplanned hysterectomy, and admission to intensive care) as well as chronic disease (gestational hypertension, gestational diabetes, eclampsia). Our primary outcome measures are the sum of these conditions for each mother, though we also examine whether each mother has any one of these conditions. For maternal morbidity and maternal transfer rates, we start our analysis in 2009 due to changes in the information provided on birth certificates over time.

We supplement county-level measures of birth outcomes with county-level population characteristics from the 5-Year American Community Survey (ACS)⁶ and county level health care resources from the Area Health Resource Files (AHRF) from 2010-2018. One limitation of the AHRF is that providers may not accurately report their practice locations. Not only do physicians fail to regularly update their addresses, AHRF (and any alternatives that track provider locations) assigns one location to a provider, even if that provider practices in multiple locations. This may be especially problematic for pinning down the location of providers in rural markets, where physicians must see patients in several locations to support a full-time practice.

2.3 Summary Statistics

Table 1 presents summary statistics for all merger transactions from 2007-2019 that involve at least one rural hospital being acquired. In 333 transactions, there are 572 hospitals that are acquired.⁷ Some transactions involve many individual hospitals; on average, a transaction that results in the acquisition of at least one rural hospital involves 22 hospitals and over

⁶We assign each 5 year survey to the final year that is included in that survey. Relatively few rural counties are included in the 1 Year ACS.

⁷Some of these hospitals are acquired multiple times. We see 499 unique hospitals acquired during this time window.

3000 beds. The combined system sees over 15,000 births a year.

We additionally present descriptive statistics for transactions where at least one of the acquirers is in the same hospital referral region (HRR) as the target (column 2), transactions where no acquirers are in the same HRR as the target, but at least one acquirer is in the same state (column 3), and transactions where no acquiring party is in the same state as the target (column 4).⁸ The bulk of our transactions and acquired rural hospitals are within the same HRR. These transactions tend to be slightly larger on average, as measured by the total number of hospital beds (or annual deliveries) involved across the involved hospitals.

We also classify mergers by the ownership structure of the involved parties, distinguishing among mergers involving entirely nonprofit parties, entirely for-profit parties, having a for-profit acquirer with a nonprofit target, and having a nonprofit acquirer with a for-profit target. The majority of our mergers involve all nonprofit parties.

Our analysis is based on either: (1) the first time a hospital was acquired in our sample period or (2) the first time that a county had a hospital located in it that was acquired in our sample period. In the case of rural hospitals, this is not very restrictive; of our acquired hospitals, 85% of the hospitals were acquired only once. Of the rural counties that experience a hospital being acquired, 74% experience only one acquisition. These distributions are shown in Appendix Figure C2. Moreover, because many rural counties contain only a single hospital (79% of all counties that have at least one hospital), these counties also only experience a single acquisition during our sample period.

Table 2 presents summary statistics at the hospital level for characteristics at the start of our sample period (2007). We characterize the differences between urban hospitals and rural hospitals in columns 1 and 2, then characterize the difference between our treated hospitals (those that are acquired at least once between 2007 and 2019) and our control hospitals

⁸A hospital referral region (HRR) is a geographic area where most of the patients in that area are referred for major cardiovascular surgical procedures and for neurosurgery. See <https://data.dartmouthatlas.org/downloads/methods/geogappdx.pdf>.

(those that were not involved in merger activity between 2007 and 2019). Column 5 reports the differences between column 3 and 4 along with stars that indicate statistical significance.

As compared with rural hospitals that have no merger activity (column 4), acquired rural hospitals (column 3) differ in their ownership structure: they are more likely to be part of a system, less likely to be government hospitals, and more likely to be private – either for-profit or nonprofit. They do not differ in profitability or in the likelihood of closure during our sample period, but are more likely to have obstetric care at baseline and are larger across a number of dimensions.

Table 3 shows summary statistics at the county level for characteristics in the NVSS in 2009, which is near the beginning of our sample.⁹ Columns (1) and (2) highlight the differences between urban and rural counties. Urban counties have substantially more births, lower transfer rates, and more women delivering in their county of residence. Birth characteristics look more similar across urban and rural counties, with the exception of anesthesia which is more common for women who live in urban counties.

Column (3) shows characteristics of rural counties with hospitals. They have more births and women are more likely to deliver in their county of residence. Column (4) shows characteristics at our treated counties, those with a hospital that is acquired from 2007-2019, and column (5) shows characteristics in the broadest control group, rural counties without an acquired hospital. Column (6) reports the differences. Our treated counties have more births, a higher fraction of births covered by Medicaid, and a higher fraction of births in the mother’s county of residence. Women in treated counties were more likely to get the recommended number of prenatal care visits, less likely to get anesthesia, and have a lower rate of chronic maternal health conditions.

To increase comparability, we also restrict attention to counties that have a hospital at some point in the sample period. Column (7) shows characteristics at counties with a

⁹2009 is the first year there are data for all the outcomes we examine in the NVSS.

hospital, but where no acquisition occurred during our sample window. Column (8) reports the differences between the treated counties and the alternative control group in column (7). Some, but not all of these differences shrink if we instead compare our treated counties to those that have a hospital at some point in our sample window.

2.4 Descriptive Analysis

We now describe the patterns in our data for rural hospital acquisitions, obstetric unit closures, and overall hospital closures, focusing in particular on differences in obstetric unit closures for hospitals that were acquired versus those that were not.

Figure 1 shows the number of annual acquisitions, obstetric unit closures, and hospital closures for rural hospitals between 2007 and 2019. During this time period, both hospital acquisitions and obstetric unit closures were common. In 2007, there were 2,021 rural hospitals, and 62.5 percent offered obstetric care; in 2019, there were 1,799 rural hospitals, and 53.0 percent offered obstetric care. There are between 15 and 35 obstetric unit closures a year and between 23 and 59 acquisitions of rural hospitals.

There is a positive association between acquisition and obstetric closure. Among all rural hospitals with obstetric units, the unconditional probability that they close their obstetric unit is 2.1 percent. Among recently acquired hospitals (in the last 5 years of our sample), the closure rate increases to roughly 3.5 percent.

Many obstetric unit closures occur independent of hospital consolidation activity; these obstetric closures have different characteristics than those at hospitals that have been recently acquired. Figures 2 and 3 characterize how closures following acquisition may differ from closures at independent hospitals. Figure 2 is a binned scatter plot that shows how the probability of obstetric unit closure is associated with the number of births in the prior year (subfigure (a)) and the distance to the next nearest hospital with obstetric services (subfigure (b)), separately by whether the hospital has been recently acquired (shown in red) or not

(shown in blue). Each bin represents 5% of hospital-year observations. We restrict our sample to hospitals that had an open obstetric unit in the previous year.

Subfigure (a) shows that for both acquired hospitals and those that were not, there is a negative relationship between birth volumes and closures, but this relationship is stronger for acquired hospitals. Subfigure (b) shows that acquired hospitals are more likely to lose their obstetric units when there is an open obstetric unit nearby. We do not find evidence of a similar relationship among hospitals that were not acquired.

In Figure 3, we compare three types of hospitals that closed obstetric units: (1) those that were not acquired and the hospital remained open, (2) those that were acquired in the five years before the obstetric unit closure, but the hospital itself remained open, and (3) those where the entire hospital closed. We examine four important hospital characteristics: the number of obstetric beds, the total number of births at the hospital, the total number of Medicaid discharges, and the hospital's profit margin, defined as $(\text{income} - \text{total cost}) / \text{income}$. We find meaningful differences for all of these among the three types of obstetric closures.

Subfigure (a) shows that closed hospitals had more beds than those that remained open. Among surviving hospitals, acquired hospitals close larger units relative to those that are closed at non-acquired hospitals. A similar pattern emerges for the number of births (subfigure (b)) and the number of Medicaid discharges (subfigure (c)). The pattern is different for hospital profit margins (subfigure (d)). Here, among surviving hospitals, differences between non-acquired and acquired hospitals are relatively small, although acquired hospitals have lower margins. This suggests acquired hospitals may respond to financial distress by shutting down units. However, we do not see selection into acquisition based on lower margins (see Appendix Figure C1 and Table B1). Note that there is substantial evidence that hospitals that shut down the entire facility were in severe financial distress immediately prior to the closure.

3 Empirical Strategy

3.1 Econometric Specification

We use a differences-in-differences research design to estimate the causal effects of hospital acquisition on hospital outcomes, leveraging variation in the timing of hospital acquisition. Our setting poses some econometric challenges. We have staggered timing and expect time-varying effects as hospitals adjust their service line offerings over time in response to the acquisition. A recent literature identifies bias in standard two way fixed effects research designs in the presence of such phenomena (for an overview of this literature, see [Baker et al., 2026](#)). [Callaway and Sant’Anna \(2021\)](#) (CS) have developed an estimator that provides unbiased estimates in settings with staggered timing and time-varying effects. We use the CS estimator and aggregate the estimated treatment effects to get interpretable estimates of the effects of the acquisition. We begin by estimating group-time specific average treatment effects:

$$ATT^{CS}(g, t) = \frac{1}{N_g} \sum_{i=1}^N I[G_i = g](Y_{i,t} - Y_{i,g-1}) - \frac{1}{N_u} \sum_{i=1}^N I[G_i = \infty](Y_{i,t} - Y_{i,g-1}) \quad (1)$$

where $Y_{i,t}$ is the outcome for hospital i in year t , G_i is the year in which hospital i is first acquired during our sample period (the hospital’s group), N_g is the number of treated hospitals in group g , and N_u is the number of untreated hospitals. We define treated hospitals as rural hospitals that are acquired during our sample period and control hospitals as rural hospitals that are not involved in any merger activity during our sample period.

We then take the weighted average of these estimated treatment effects to obtain the treatment effect in each year relative to the merger:

$$ATT_k^{ES} = \sum_g w(g, k) ATT^{CS}(g, g + k),$$

where the weights $w(g, k)$ depend on the relative sizes of the groups representing acquisitions in each year. We refer to these as our event study estimates, as they are analogous to traditional event study estimates.

We additionally report the simple average of all of our post-treatment ATTs described above. We refer to these as our difference-in-difference estimates. We cluster standard errors at the hospital level. We report simultaneous confidence intervals that are robust to multiple hypothesis testing from [Callaway and Sant’Anna \(2021\)](#).

We estimate similar models using county-level outcomes as the dependent variable to examine the effects of hospital acquisition on county health care resources and patient outcomes. In these regressions, the treatment is whether there was an acquired hospital located in the county. The interpretation of the identified parameter differs from our main hospital-level regressions in an important way: the parameter in the hospital-level regressions reflects the direct effect of the acquisition on the hospital, while the parameter in the county-level regression reflects the effects on county-level resources and outcomes for county residents. These, in turn, may reflect changes at the targeted hospital, responses by other providers inside and outside the affected county, and choices of county residents.

The key identifying assumption is that hospitals (or counties) would have trended similarly in the absence of treatment (parallel trends). Note that this assumption does not require that hospitals in our treatment and control group are similar, as we have evidence from [Table 2](#) that our treated hospitals are different at the beginning of our sample period across several characteristics. To most directly assess the plausibility of this assumption, we examine the event study estimates for the years leading up to the acquisition.

In Appendix [A.3](#), we discuss a number of further analyses to assess the validity of this assumption: we conduct formal pre-trends testing, examine whether there is evidence of differential pre-trends in a broad set of hospital characteristics, test whether there is selection on observable characteristics into acquisition, and attempt to predict acquisition. We find no evidence contradicting parallel trends.

While we view the results of these exercises as consistent with the parallel trends assumption for hospitals, there are some specifications where we see some evidence of non-parallel pre-trends when we examine outcomes at the county level. For this reason, we estimate synthetic difference-in-difference models using the estimator proposed by [Arkhangelsky et al. \(2021\)](#), following the implementation in [Clarke et al. \(2024\)](#) and [Ciccia \(2024\)](#). This estimator combines the properties of synthetic control and difference-in-difference estimators and re-weights never-treated hospitals (counties) to match outcome trends in the pre-treatment period in treated hospitals (counties).

One limitation of this method is that it requires a balanced panel. For county-level outcomes, we already create a balanced panel to remove concerns about changes in data reporting as a result of the merger. However, at the hospital level this restriction requires us to drop all hospitals that close or drop out of AHA survey reporting. For that reason, the CS estimator remains our primary specification when examining hospital level outcomes.

3.2 Robustness

We additionally assess robustness to a variety of alternative specification choices. First, our main CS estimator uses only untreated hospitals in the control group; we can instead include the “not yet” treated hospitals in our control group. Untreated hospitals are those that are not involved in a merger anytime during our study period. “Not yet” treated hospitals are hospitals that have not been acquired up until the relevant year, but are acquired at some point in the future. This broadens the control group in order to examine whether the

estimates are sensitive to delineation of the control group.

Second, we can exclude all “never treated” hospitals, leveraging only comparisons between the treated hospitals and the “not yet” treated hospitals, in order to specifically use variation that only comes from the timing of acquisitions, rather from whether or not a hospital is acquired. Third, since the treated hospitals are different at baseline than those in the control group, we estimate a model that uses inverse probability weights to match on pre-period hospital characteristics, specifically hospital margins, number of general adult beds, number of births, total inpatient days, and payroll expenses.

We conduct additional sensitivity analysis to our classification of treatment and control groups. In our main specification, treated hospitals are those that are acquired at least once in the period between 2007 and 2019 and control hospitals are those that have no involvement in merger activity in that time period. We do not restrict treated hospitals to not also be involved in merger activity as acquirers during the sample period. Thus, some hospitals in our treatment group may not only be acquired, but may also be acquirers as well, which may potentially contaminate the estimates of the impacts of being acquired.¹⁰ Alternatively, we could drop all hospitals from our sample who are part of acquiring systems at any time, or include hospitals that are part of acquiring systems in both the treatment and control groups.

There are advantages and disadvantages to all three approaches. Our main specification allows us to estimate the effect of the full set of acquisitions, but has an asymmetry between our treatment and control group. Restricting the treatment group to only hospitals that are never part of acquiring systems removes 76% of acquisitions from our sample. Including acquiring hospitals in both the treatment and control groups complicates the interpretation of the estimates.

¹⁰76% of our target hospitals are additionally part of an acquiring system in another transaction during our sample period.

When examining county level outcomes, our main specification considers treated rural counties where a hospital was acquired and untreated rural counties where a hospital was not acquired, regardless of other merger activity. We also estimate models that drop from our sample counties that contain any hospitals that are acquirers. Finally, to improve comparability between our treatment and control groups, we estimate models that restrict the sample to only rural counties that have a hospital in at least one year in our sample period. We report the results from these robustness exercises in the section that follows.

4 Results

In this section, we present event study estimates examining the impacts of rural hospital acquisitions on obstetric services availability, births, resource availability, and patient outcomes.

4.1 Hospital Level Analysis: Obstetric Care Units and Deliveries

We begin by estimating the effect of hospital acquisition on obstetric units. Figure 4 shows that acquisition reduces the probability that the hospital has an obstetric unit, with the effect building over time.¹¹ 5 years after the acquisition, hospitals are 11% less likely to offer obstetric care. As is clear from the figure, the magnitude of the effect increases over time. This makes sense, as adjustments take time to plan and implement after an acquisition. We see no evidence that acquired hospitals were trending differently to non-acquired hospitals in the pre-period.

Column 1 of Table 4 shows the estimate of the average treatment effect on the treated

¹¹Note that we estimate this unconditionally. That is, we do not restrict our sample to hospitals that have an obstetric unit in the pre-period. We do this for two reasons. First, we want to allow for the possibility that acquisitions could lead to either the opening or closure of an obstetric unit. Second, because acquisitions occur at different time periods throughout the sample, there is not a clean comparison group to hospitals that have an obstetric unit before acquisition.

(ATT) from our main CS specification. The ATT estimate is -0.05, which is roughly a 9 percent decrease in the probability of having an obstetric unit on average across all years post-acquisition.

As discussed above, we assess robustness to several possible alternative specification choices. First, in column 2, we include not-yet-treated units in our control group. Column 3 drops all never-acquired hospitals from the analysis. This specification leverages variation in the timing of mergers, which substantially reduces our sample size. Column 4 conditions on pre-period hospital characteristics using the doubly-robust method suggested in [Callaway and Sant’Anna \(2021\)](#) using inverse probability weighting to model selection into treatment. Column 5 reports estimates from a synthetic difference-in-difference model, following [Arkhangelsky et al. \(2021\)](#), which we estimate to address any remaining concerns about parallel trends. The point estimates from all of these approaches are quite close in magnitude, and have the same qualitative impact, which suggests the main results are robust to the variations we examine.

In Appendix Table [B2](#), we examine robustness to various delineations of the treatment and control groups, as discussed above. There is some variation in the point estimates across the various treatment/control group definitions, but they are mostly fairly close to the point estimate from the main specification, and capture the same qualitative impact – acquisitions lead to an increased probability of obstetric unit closure.

In Appendix Table [B4](#), we report our estimated average treatment effects separately by cohort. We show corresponding event study figures in Appendix Figure [C3](#). Across most treated cohorts, we find negative estimates of the effects of acquisition, though these effects are strongest for hospitals acquired in 2010 and 2015.

Figure [5](#) presents the event study estimates of the impact of hospital acquisitions on the number of births at acquired hospitals. There is a 10% reduction in the number of births. This large reduction in births is unsurprising, given our previous results about obstetric unit

closures. In Appendix Figure C7, we show suggestive evidence that this reduction in births is predominately occurring at hospitals that shut their obstetric unit. We plot the number of births at the hospital each year before and after the merger. Hospitals that do and do not close their obstetric unit show similar pre-merger trends, but there is a sharp decline in the number of births at hospitals with obstetric unit closures, with no break in the trend at hospitals that did not close their obstetric unit.

4.1.1 Effect Heterogeneity

The effects of acquisition may depend on many factors, including the geographic proximity of the acquiring system, the characteristics of the areas where the hospitals are located, and the ownership types of the target and acquiring hospital. Table 5 reports estimates for these possible sources of heterogeneity. In each of these specifications, we examine the effects of a specific type of acquisition or for a specific kind of hospital by either varying the treatment or the included sample and estimating Equation 1 and aggregating the coefficients.

Panel A classifies acquisitions by whether the acquiring hospital¹² is located in the same HRR as the target hospital, the acquiring hospital is located in the same state but not in the same HRR, or all hospitals in the acquiring system are located across state lines. We estimate our main specification where treatment is defined as being acquired by a system that fits these classifications. Control hospitals are non-acquired hospitals. One might expect closure of a target’s obstetric unit post-acquisition to be more likely the closer the acquirer is to the target. We examine that here.¹³

The estimated coefficients are not statistically distinguishable from one another, though the largest and only statistically significant effect is for mergers involving hospitals within the same HRR. This suggests that proximity may play a role in obstetric unit closures, although

¹²Or some hospital in the acquiring system.

¹³Note that the majority of mergers involving rural hospitals are within-HRR mergers; roughly 20 percent involve hospitals in different HRRs but in the same state, and less than 10 percent involve a merger across state lines (see Table 1).

the evidence isn't strong.

Panel B classifies by whether the acquiring system has any hospitals located in an urban area (column 1) or no hospitals in an urban area (column 2), and whether the target hospital is outside of CBSAs (column 3) or in a micropolitan area (column 4).¹⁴ One might expect the effects of an acquisition to differ based on the locations of the acquirer and of the target. Rural systems may behave differently from those that consist of hospitals from both urban and rural areas. Hospitals outside of CBSAs are further from population centers and may be smaller. We find that the effects are larger when rural hospitals are acquired by rural systems and are located outside of CBSAs. Thus the effects of an acquisition on obstetric unit closure are strongest for those acquisitions that are the most rural.

Panel C classifies mergers based on the ownership structure of the acquired and acquiring parties. It seems plausible that whether a target hospital's obstetric unit is closed post-acquisition could vary depending on whether the acquirer is for-profit or nonprofit, and similarly, based on the ownership status of the target. In column 1, both parties are nonprofits ("All Non-Profit"). In column 2 both parties are for-profits ("All For-Profit"). In column 3, the acquirer is a nonprofit and the target is a for-profit ("NP Acquires FP"). In column 4, the acquirer is a for-profit and the target is a nonprofit ("FP Acquires NP"). We find statistically significant negative effects for mergers where non-profit hospitals are acquired. These effects are larger when a for-profit system is the acquiring party, as one might expect. We also find negative effects for the acquisition of for-profit hospitals, although these events are relatively rare, so we lack statistical significance.

4.1.2 Other Hospital Resources

We test for changes in other hospital service line offerings including emergency room (ER), psychiatric care, substance use care, neonatal intensive care (NICU), and pediatric intensive

¹⁴The classification of hospitals may change over time as micropolitan areas are redefined. For this analysis, we take the point in time classification of the hospital when restricting the sample

care (PICU). We examine these services because they are either frequently associated with being unprofitable (ER, psychiatric care, and substance use care) (Horwitz and Nichols (2022)) or associated with the health of babies and children (NICU and PICU). The estimates are shown in Appendix Table B5. We find small declines in the probability that acquired hospitals have emergency departments or substance use care, but no change in the other service line offerings. However, among rural hospitals, NICUs and PICUs are rare, so it is unlikely we would find evidence of impacts on these service offerings.

4.2 County Level Analyses: Health Care Resources and Patient Outcomes

4.2.1 Health Care Resources

While we have established that acquisitions lead to hospital obstetric unit closures, the total impact on available health care resources is unclear, as other health care providers may adjust in response to the closure. Other hospitals in a county may adjust their capacity in response (or there may be new entry) and physicians and midwives may change their practice locations in response to a closure. We therefore examine the net effect of acquisitions on the number of obstetric hospital beds in a county, the number of hospital obstetric units in a county, and on the numbers of physicians and nurse midwives in a county.

Figure 6 displays estimates of the effect of a rural hospital acquisition on the number of obstetric beds in the county (subfigure (a)), the number of hospitals with an obstetric unit (subfigure (b)), the number of obstetrician/gynecologists (OBGYNs) per capita (subfigure (c)), and the number of midwives per capita (subfigure (d)). We additionally report CS event study estimates in Figure C6 and a variety of robustness checks in Appendix Table B7.

Consistent with our hospital-level results, we find substantial reductions in county-level

health care resources. There is a roughly 10 percent decline in the number of obstetric beds and a 13 percent decline in the number of hospitals with obstetric units, across all years. By 5 years after the acquisition, there is a 17 percent decline in the number of beds and a 17 percent decline in the number of obstetric units. This implies that nearby hospitals are not expanding capacity in response to changes at the acquired hospital, either at all, or by enough to offset declines at the acquired hospital.

This decline in hospital resources may spill over to other obstetric resources in the county. If the number of obstetric units and obstetric beds decline, there will be fewer essential resources for obstetricians and midwives, so one would expect that some of them would choose to relocate and practice elsewhere. As expected, we find a reduction in the per capita number of obstetricians practicing in the county.¹⁵ We find no change in midwives per capita. This null result is unsurprising, given the low baseline levels of midwives; on average, there are 0.1 midwives per 10,000 people in the rural counties in our sample.

Our main specification includes all counties in the analysis. We prefer this because it does not require us to drop counties where the acquirer and target hospital are in the same county, and thereby lose statistical power and precision. In the appendix, we show alternative specifications that estimate these effects where we drop all counties that contain a hospital that is ever identified as the acquiring party in a merger transaction. We additionally estimate models where the sample is restricted to counties that have a hospital at some point in our sample period. Both of these samples are substantially smaller than the one we use for our main specification. These results are shown in Appendix Table B7. The estimates are qualitatively similar, though we lack precision and lose statistical significance in some specifications.

Because counties may not be a large enough geographic area to capture responses by

¹⁵Our data on the number of obstetricians and midwives come from the Area Health Resources File. As discussed in the data section, these measures may be problematic because they rely on self-reported location and do not account for providers who practice in multiple locations.

neighboring hospitals, we estimate similar specifications looking at hospital level resources at the level of hospital referral regions (HRR), which are substantially larger than counties.¹⁶ We restrict to HRRs that contain at least one rural hospital at some point in our sample period. We define treatment as the first time a rural hospital is acquired in the HRR during our sample window. Note that this necessitates a change in the events that are included; as we increase the geographic area that we consider treated, the requirement to consider only the first event in our sample window becomes more restrictive.

We show these estimates in Appendix Table B6.¹⁷ Column (1) examines the effects on total obstetric units. Column (2) examines the effects on total obstetric beds. We find large negative effects on both. These estimates are larger than our estimates at the county-level because acquisitions frequently affect multiple hospitals within the same HRR. Columns (3) and (4) test for offsetting effects at other hospitals in the HRR. Treatment here is defined as acquisition of another hospital in the same HRR. We find no evidence of offsetting effects in either the probability that other hospitals in the same HRR have an obstetric unit or in the number of beds.

4.2.2 Patient Outcomes and Treatments

We use data from birth certificates at the county-level to test whether acquisitions change patient outcomes. We report event study estimates from the synthetic difference-in-difference estimator in Figure 7 with aggregated ATT estimates reported in Appendix Tables B8 and B9. We first examine the probability that women deliver in their county of residence. As a baseline, 59 percent of women who live in a rural county with a hospital obstetric unit deliver within their county of residence, suggesting a preference for local deliveries. Across all rural counties, 31 percent of women deliver in their county of residence.

¹⁶There were 306 HRRs in 2007 versus over 3,000 counties.

¹⁷We do not include OBGYNs or midwives here as our data on these providers exist at the county level, which does not cleanly map to HRRs.

Figure 7 subfigure (a) reports synthetic difference-in-difference estimates of the effect of a hospital acquisition in the mother’s county of residence on where she delivers. We estimate a 3 percentage point reduction (10 percent reduction relative to the outcome mean across all counties) in the probability that mothers deliver in the county in which they reside. The bulk of this reduction comes from the closure of obstetric units; Appendix Figure C8 shows the time series of the probability that mothers deliver in the county in which they reside for counties where a rural hospital was acquired, split by whether the hospital closed its unit. There is no break in the trend for counties that did not see an obstetric unit closure post-acquisition.

In subfigure (b), we show that this reduction in in-county births results in a small increase in distance traveled of 0.6 miles, as measured by the distance between the county of residence and the county in which the baby is born.¹⁸ While this effect seems small in aggregate, if we scale this by the percent of women who are displaced, we find substantially higher estimates: women who are displaced are traveling roughly 20 miles further.

Estimating birth displacements is difficult in this context because of the size disparity between acquired and acquiring hospitals. Rural hospitals that are acquired have 285 births a year. In contrast, acquiring hospitals with rural targets see 1,170 births per year on average. To complicate matters, we are unable to identify births at specific hospitals in the NVSS data. Given these caveats, we calculate the change in births in the county of the nearest acquiring hospital with an obstetric unit for each treated hospital by simply comparing the percentage delivering in this county before and after the acquisition. The time series is shown in Figure 8. We find that on average, births in the county of the nearest acquiring hospital with an obstetric unit rise just over 1.5 percentage points, or 9 percent. This is suggestive evidence that many births are displaced to the acquirer, as we see about a 3 percentage point

¹⁸Note that we winsorize the distance traveled to minimize the influence of outliers where women deliver across the country from their primary residence.

change in births in the mother’s county of residence, the location of the target hospital.

Subfigures (c) and (d) evaluate the effects of acquisition on prenatal care. We find decreases in the probability that prenatal care starts in the first trimester, consistent with the decrease in OBGYN availability we find. We additionally estimate small decreases in the probability that women receive the recommended number of prenatal visits.

Subfigures (e)-(g) evaluate the effects of acquisition on interventions during labor. We find no evidence of changes in either c-section or induction rates. For c-section rates, we can rule out effects larger than about 1.5% in either direction. For inductions, we can reject decreases larger than 1.4% or increases larger than 3.5%. However, we do find increases in the use of anesthesia. This increase in anesthesia could reflect either differences in the availability and ease of access or differences in practice style across hospitals. In subfigure (h), we find small declines in the probability that a mother is transferred during labor. We take this as evidence that women are delivering in more resourced hospitals.

Subfigures (i) - (l) evaluate the effects of acquisition on preterm labor and maternal health. We find very small increases in the probability of preterm labor, though these estimates are not consistent across alternative specification choices. We find no evidence of changes in maternal morbidity, measured a variety of ways. We reject decreases in severe maternal morbidity of more than 4.2% or increases larger than 8.8%, decreases in chronic maternal morbidity of more than 1.3% or increases of more than 8.0%, and decreases in overall maternal morbidity of more than 1.4% or increases of more than 7.2%.

In these subfigures, we report the effects on indices that sum across reported conditions; in Appendix Table [B9](#), we additionally report estimates from regressions that examine whether women are reported to have any condition within the category. We similarly find no evidence of effects.

Our preferred specification is the synthetic differences-in-differences approach. Our sample is naturally balanced, making this model a good fit for our setting. We additionally

estimate a variety of Callaway Sant’Anna specifications and show robustness to a number of alternative specification choices, including those where we include observations not yet treated in our control group, vary the sample to drop counties where acquiring parties are located, and restrict to counties that contain a hospital. These results are shown in Appendix Tables B8 and B9, with corresponding event study plots in Appendix Figure C9.

4.2.3 Infant and Maternal Mortality

Figure 9 presents our estimates of the effects of acquisitions on infant and maternal mortality. Subfigure (a) reports estimates of the effect on infant mortality. Subfigures (b) reports estimates for the effect on maternal mortality. In subfigures (c) and (d), we estimate effects for two high-risk groups, Black women and women of advanced maternal age (AMA).¹⁹ We narrow in on these populations as their baseline rates of maternal mortality are higher, increasing our power to detect changes in mortality. Across our four measures of mortality, we find little evidence of an acquisition effect, though there is some suggestive evidence that Black maternal mortality declines. This decline is not evident in alternative specifications, shown in Appendix Table B10. Generally, our point estimates are imprecisely estimated, with wide confidence intervals.

4.3 Are Acquisition-Induced Closures Like Other Closures?

To our knowledge, we are the first to consider the effects of rural hospital acquisitions on maternal health outcomes. However, an existing literature studies the effects of rural obstetric closures more broadly. We find important differences between our results and that of the prior literature, which makes clear that acquisition-induced closures can be a very different phenomenon than other closures. In Table 6, we compare our results to those of

¹⁹Note that this estimator requires a balanced panel so our effects are estimated only on counties that have at least one birth among these sub-populations for every year in our sample.

[Fischer et al. \(2024\)](#) and [Battaglia \(2025\)](#), the two papers in the literature that study rural hospital obstetric closures in a setting similar to ours. There are a few key differences. While we find similar effects on prenatal care, we find no effects on gestational age and no changes in induction rates or c-section rates. We additionally find no effects on maternal health.

There are a few reasons why the effects of acquisitions may be different from those of closures generally. Acquisitions may induce quality changes at hospitals that keep their obstetric unit open, systems may better be able to absorb displaced patients, and the shuttered units may be different than those at hospitals that closed completely. As shown in [Figure 2](#), closures following acquisitions are at hospitals that are larger on average than hospitals that independently close their obstetric unit and are at hospitals that are closer to the next nearest obstetric unit.

4.4 Alternative Explanations

It is possible that acquired hospitals may face different demographic trends than those that are not, leading to biased estimates. For instance, hospitals in areas with declining populations or sagging economies may be more willing to entertain takeover offers from hospital systems. Reductions in deliveries, closures of obstetric units, and poorer birth outcomes might be due to changing demographics in our treated counties, rather than takeovers. To test for this, we run our event study specification examining demographic characteristics of the county.

In particular, we look at the $\log(\text{population})$, median income, fraction of the population between 18-34, fraction of the population white, and fraction uninsured. We report our synthetic difference-in-difference estimates in [Table 7](#), with additional specifications shown in [Appendix Tables B11](#) and [B12](#). We show corresponding event study figures in [Appendix Figure C10](#).

In [Panel A](#), we find no evidence that counties with acquired hospitals were trending dif-

ferently in population size, income, or race. We find that post-acquisition, the percent of the population 18-34 declines, but this effect is small (approximately 1 percent change). Thus, we think it is unlikely that our results are being driven by changes in county demographics in response to the acquisition.

In Panel B, we further examine the age, education, race and Medicaid status of mothers who live in the county of acquired hospitals. We find no evidence that maternal age or Medicaid status are changing. There is a small increase in the percent without a high school diploma and a small decrease in the percent white.

It is also possible that acquisitions change the probability of (total hospital) closures. This would cause problems for our main hospital-level regressions since closed hospitals in the control group would drop out of our sample over time. To address this, we estimate the effect of hospital acquisition health care resources at the county level as in Table B7. Because counties do not drop out of our sample, if the alternative was an entire hospital closure, this change in resources would be captured in our county-level estimates. We additionally re-estimate these specifications excluding all counties with hospital closures; our estimates are in line with our preferred specifications (see Appendix Table B13), thereby alleviating concern about confounding due to acquisitions changing the probability of hospital closure.

5 Summary and Conclusions

While there has been extensive research on the impact of hospital acquisitions on prices and some research on quality, there is a paucity of evidence of the impact on access. This is especially important in the case of rural hospitals and particularly with regard to obstetric care. We provide evidence on the impacts of acquisitions on access to obstetric care at targeted rural hospitals and additional evidence of impacts on quality.

We find that while hospital acquisitions increase the likelihood of closures of obstetric

units and reductions in local health care resources, there is mixed evidence of the effects on patients. In counties where a hospital is acquired, we find increased use of procedures that are suggestive of more-resourced care, but declines in prenatal care. We find no evidence of changes in maternal morbidity or mortality.

Our findings have important implications for policy toward rural hospital acquisitions. The fact that some acquisitions lead to obstetric unit closures, and prior research shows that obstetric unit closures harm access and quality, could add to skepticism towards these acquisitions. Policy makers might thus be tempted to require acquirers to continue to operate obstetric units at target hospitals as a means of maintaining access for rural communities.

We find, however, that acquisition-induced closures do not consistently result in poorer outcomes. Our findings suggest that requiring acquirers to keep obstetric units open at acquired rural hospitals may not achieve the desired results, and might even prove counterproductive. Requiring that they do so may discourage acquisitions, including those that would ultimately benefit patients at the acquired hospitals. Given this nuanced picture, policy makers need to consider approaches that provide acquirers with flexibility while assuring access to safe, quality care for the patient population of the acquired hospital.

There is substantially more work to be done examining the impacts of mergers on patient access. We hope our work inspires new research addressing this understudied and important issue.

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6 Tables

Table 1: Acquisition Summary Statistics

	All Transactions (1)	Within HRR (2)	Across HRR (3)	Across State (4)
Total Transactions	333	239	60	34
Total Rural Targets	572	439	80	53
All Non Profits	178	132	35	11
All For Profits	25	15	4	6
For Acquires Non	55	39	8	8
Non Acquires For	15	13	1	1
Total Involved Hospitals	21.589	23.192	16.983	18.441
Total Beds	3314.802	3619.954	2199.667	3137.647
Total Births	15557.14	17046.46	9800.634	15246.65

Notes: This table reports summary statistics for all merger transactions from 2007-2019 involving at least one rural target. “All Non Profits”, “All For Profits”, “For Acquires Non”, “Non Acquires For” indicate the ownership composition of the transaction with For denoting for-profit hospitals and Non denoting nonprofit hospitals. If any hospitals in the acquiring system are identified as for-profit hospitals, we classify the acquiring system as for-profit. If any hospitals acquired as part of the transaction are nonprofit, we classify the transaction as the acquisition of nonprofits. The remaining transactions involve government hospitals.

Table 2: Hospital Summary Statistics

	Urban	Rural	Rural, Acquired	Rural, No Merger Activity	Difference, Acquired v. No Merger
In System	0.636	0.416	0.436	0.074	0.362***
Government	0.146	0.363	0.270	0.527	−0.257***
For Profit	0.212	0.106	0.145	0.043	0.103***
Non Profit	0.642	0.530	0.585	0.430	0.155***
Margin	0.025	0.022	0.018	0.011	0.007
Has Obstetric Care	0.735	0.625	0.672	0.557	0.115***
Medicaid Discharges	2114.947	417.632	510.068	299.753	210.316***
Total Admissions	11339.618	2295.037	2741.118	1713.804	1027.314***
Total Births	1365.765	243.939	285.409	180.304	105.105***
Closure 2007 - 2019	0.068	0.063	0.066	0.075	−0.008
N	2660	2021	482	869	

Notes: This table reports 2007 characteristics for hospitals in the AHA data. Column (1) reports averages for all urban hospitals. Column (2) reports averages for all rural hospitals. Column (3) reports averages for all rural hospitals acquired at least once from 2007–2019. Column (4) reports averages for all rural hospitals involved in no merger activity from 2007–2019. Column (5) reports the difference between columns (3) and (4).

Table 3: County Level Summary Statistics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Urban	Rural	Rural, w/ hospital	Rural, hosp ac- quired	Control (all)	Difference (4)–(5)	Control (has hospi- tal)	Difference (4)–(7)
Births	3239.125	339.822	394.310	534.166	284.140	250.026***	340.927	193.240***
Medicaid	0.346	0.432	0.432	0.456	0.425	0.031**	0.422	0.034**
Delivery in County of Residency	0.467	0.325	0.398	0.434	0.294	0.140***	0.384	0.050***
Distance	15.296	27.566	24.511	21.120	29.413	−8.293***	25.805	−4.685***
C Section	0.323	0.327	0.327	0.330	0.326	0.005	0.325	0.005
Induction	0.264	0.288	0.288	0.289	0.288	0.001	0.287	0.001
Anesthesia	0.439	0.416	0.417	0.383	0.426	−0.042***	0.430	−0.047***
Mother Transferred	0.004	0.010	0.010	0.009	0.010	−0.001	0.010	−0.001
Prenatal Care in First Trimester	0.720	0.711	0.714	0.717	0.709	0.008	0.713	0.004
Recommended Prenatal Visits	0.514	0.462	0.471	0.499	0.452	0.047***	0.460	0.038***
Preterm	0.120	0.123	0.122	0.125	0.123	0.002	0.121	0.003
Maternal Morbidity Index	0.066	0.074	0.075	0.067	0.076	−0.009**	0.078	−0.011***
Severe Maternal Morbidity Index	0.008	0.011	0.012	0.010	0.011	−0.001*	0.012	−0.002***
Chronic Maternal Morbidity Index	0.057	0.063	0.064	0.057	0.065	−0.007**	0.066	−0.009***
N	1057	2079	1676	463	1616		1213	

Notes: This table shows summary statistics at the county level for 2009. Distance is measured in miles from the county centroid of the mother’s home county to the county in which she delivers. First trimester prenatal care is defined as the first prenatal care appointment occurs within the first three months of pregnancy. Recommended care indicates whether the mother attended at least 12 prenatal visits. The severe maternal morbidity index includes maternal transfusion, perineal laceration, ruptured uterus, unplanned hysterectomy, and admission to intensive care. The chronic maternal morbidity index includes gestational hypertension, gestational diabetes, eclampsia. The overall maternal morbidity index includes both sets of conditions. Data Source: NVSS

Table 4: Difference-in-Difference Estimates of the Effects of Hospital Acquisition on Obstetric Units

	(1) Main CS	(2) Include Not Yet Treated	(3) Drop Never Treated	(4) IPW	(5) Synthetic DiD
ATT	-0.0494*** (0.0140)	-0.0496*** (0.0139)	-0.0565*** (0.0165)	-0.0478*** (0.0149)	-0.0549*** (0.0150)
N	16615	16710	5983	15918	14742
Outcome Mean	.55	.55	.61	.59	.57

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table presents our estimated aggregated average treatment effects for the effect of hospital acquisition on obstetric units. Column 1 reports our baseline estimate that does not include the not-yet-treated in the control group. In column 2, the control group includes both never-treated hospitals and not-yet-treated hospitals. Column 3 estimates this model using only hospitals that are acquired at some point in the sample period. Column 4 implements inverse probability weighting on pre-period hospital characteristics. Column 5 reports estimates from a synthetic differences-in-difference model. Standard errors are clustered at the hospital level.

Data source: AHA Annual Survey of Hospitals

Table 5: Effect Heterogeneity

Panel A: Geographic Distance				
	(1)	(2)	(3)	
	Within HRR	Across HRR	Across State	
ATT	-0.0592*** (0.0196)	-0.0183 (0.0219)	-0.0425 (0.0369)	
N	21887	20339	19729	
Number Acquired	266	124	76	
Outcome Mean	.57	.57	.57	
Panel B: Acquirer and Rural Characteristics				
	(1)	(2)	(3)	(4)
	Urban	No Urban	Outside CBSA	Micro CBSA
ATT	-0.0366*** (0.0130)	-0.165** (0.0658)	-0.0651*** (0.0211)	-0.0356** (0.0167)
N	24213	19761	14458	9974
Number Acquired	421	47	276	271
Outcome Mean	.58	.56	.43	.79
Panel C: Ownership				
	(1)	(2)	(3)	(4)
	All Non-Profit (NP)	All For-Profit (FP)	NP Acquires FP	FP Acquires NP
ATT	-0.0307* (0.0180)	-0.0191 (0.0288)	-0.183 (0.121)	-0.119** (0.0528)
N	21680	19731	19374	19915
Number Acquired	204	50	12	119
Outcome Mean	.58	.57	.57	.57

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table presents our estimates of the aggregated treatment effect of hospital acquisition on obstetric units for different types of mergers and hospitals. Panel A classifies mergers by geographic distance. Within HRR denotes mergers where the target and at least one hospital in the acquiring system are located in the same HRR. Across HRR denotes mergers where the target and no acquiring hospitals are located in the same HRR, but at least one hospital in the acquiring system is located in the same state. Across state denotes mergers where the target hospital is located in a different state from all hospitals that make up the acquiring system.

Panel B classifies acquisitions by whether the acquiring system has any hospitals located in an urban area (column 1) or no hospitals in an urban area (column 2), and whether the target hospital is outside of CBSAs (column 3) or in a micropolitan area.

Panel C classifies mergers by the ownership structure of the target and acquirer. Column 1 examines the effects of mergers where both the target and acquiring system are non-profits. Column 2 examines the effects of mergers where both the target and acquiring systems are for-profits. Column 3 examines the effects of mergers where the target is a for profit hospital and the acquiring system is non-profit. Column 4 examines the effects of mergers where the target is a non-profit hospital and the acquiring system is for-profit.

Data source: AHA Annual Survey of Hospitals

Table 6: Comparison with Obstetric Closure Literature

Outcome	Effect of Acquisition	Effect of Obstetric Closure (Fischer et al. (2024) and/or Battaglia (2025))
Prenatal Care	Decrease	Decrease
Gestational Age	No Effect	Decrease
Inductions	No Effect	Increase
C-Sections	No Effect	Decrease
Infant Mortality	No Effect	No Effect
Maternal Morbidity	No Effect	Decrease

Table 7: Synthetic Difference-in-Difference Estimates of the Effects of Hospital Acquisition on Maternal and County Characteristics

<i>Panel A: County characteristics</i>					
	Log(Population)	Median Income	% Pop 18–34	% Pop White	% Uninsured
ATT	-0.000444 (0.00191)	-189.9 (156.0)	-0.00136** (0.000627)	-0.00164 (0.00118)	-0.00191 (0.00144)
N	21912	21901	21912	21912	12712
Outcome Mean	9.64	43522.52	.2	.85	.13
<i>Panel B: Maternal characteristics</i>					
	Medicaid	Maternal age	White	College+	Less than high school
ATT	0.0128 (0.0107)	-0.00639 (0.0211)	-0.00637*** (0.00165)	0.00382 (0.00376)	0.0152*** (0.00432)
N	18029	26611	26611	26611	26611
Outcome Mean	.47	26.66	.86	.17	.14

Standard errors in parentheses

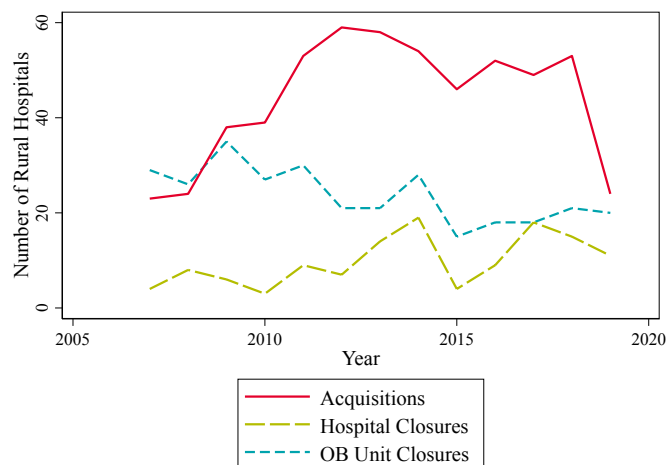
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports estimates for the effect of acquisition on county characteristics (Panel A) and on maternal characteristics (Panel B).

Data Sources: U.S. Census Bureau American Community Survey, NVSS

7 Figures

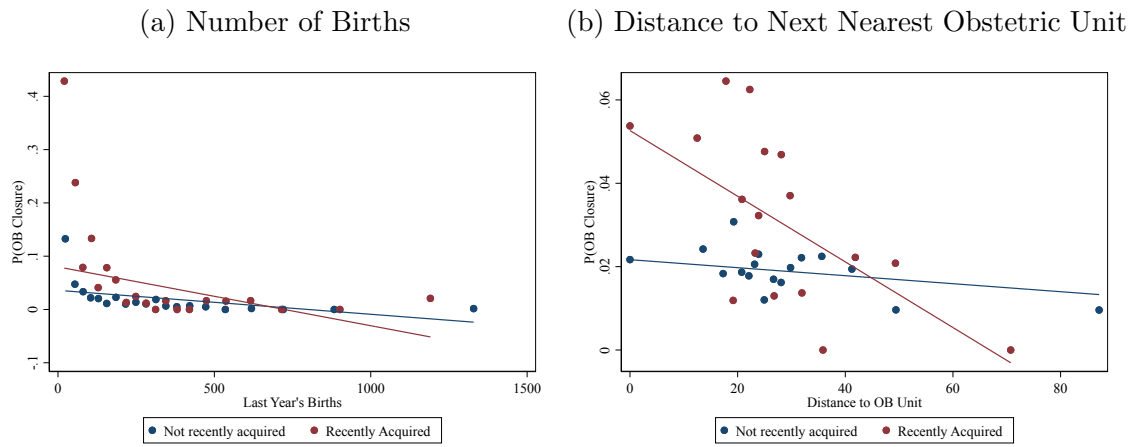
Figure 1: Acquisitions, Obstetric Closures, and Hospital Closures at Rural Hospitals



Notes: This figure shows the annual number of rural hospitals that are acquired, closed their obstetric units, or closed entirely from 2007-2019.

Data source: AHA Annual Survey of Hospitals

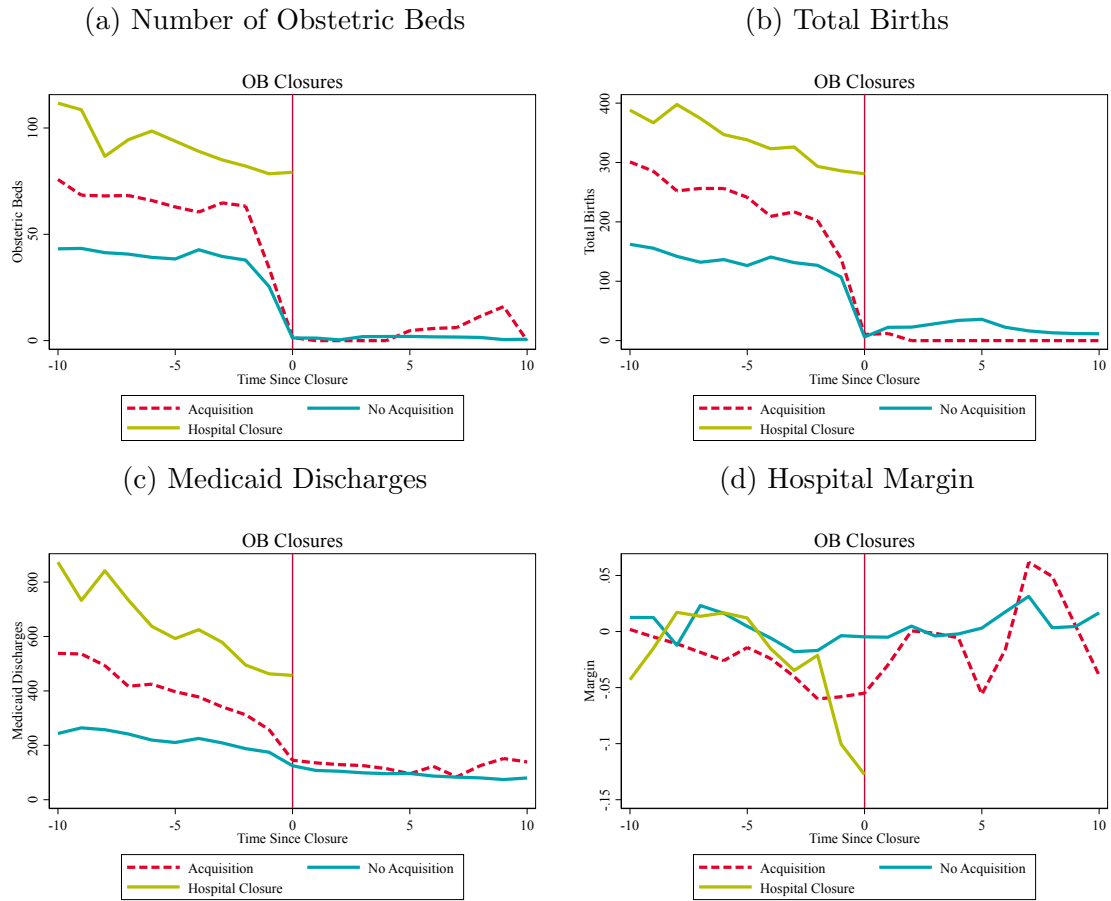
Figure 2: Probability of Closure by Acquisition Type



Notes: This figure presents binned scatter plots that show how the probability of obstetric unit closure varies with the number of births and acquisition status (subfigure (a)) and with the distance to the nearest obstetric unit and acquisition status (subfigure (b)). Each point represents 5% of hospital-year observations. Only hospitals with an obstetric unit open in the previous year are included.

Data source: AHA Annual Survey of Hospitals

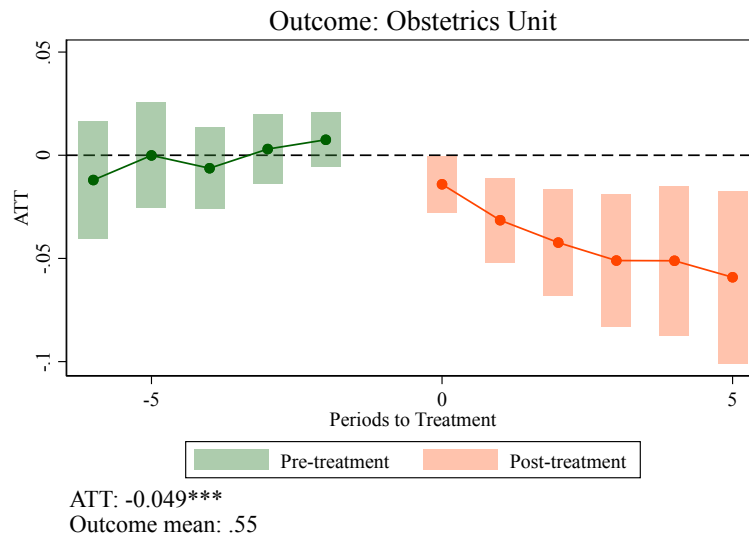
Figure 3: Comparison of Obstetric Closures



Notes: This figure shows the time series of characteristics of hospitals over time leading up to the closure of their obstetric unit. Subfigure (a) looks at the number of obstetric beds. Subfigure (b) looks at the number of births. Subfigure (c) looks at the average number of Medicaid discharges. Subfigure (d) examines hospital margins.

Data source: AHA Annual Survey of Hospitals

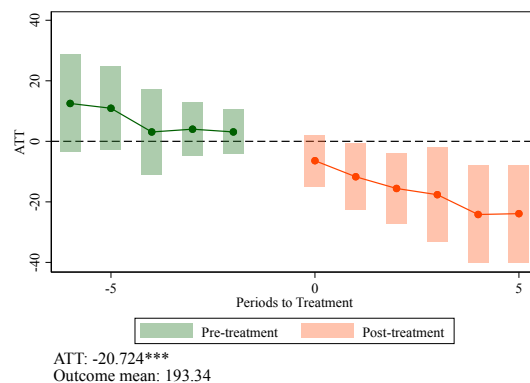
Figure 4: Effect of Acquisition on Probability of Hospital Obstetric Unit



Notes: This figure reports estimates of the effect of hospital acquisition on operating an obstetric unit. These are calculated using the [Callaway and Sant'Anna \(2021\)](#) event study estimator.

Data source: AHA Annual Survey of Hospitals

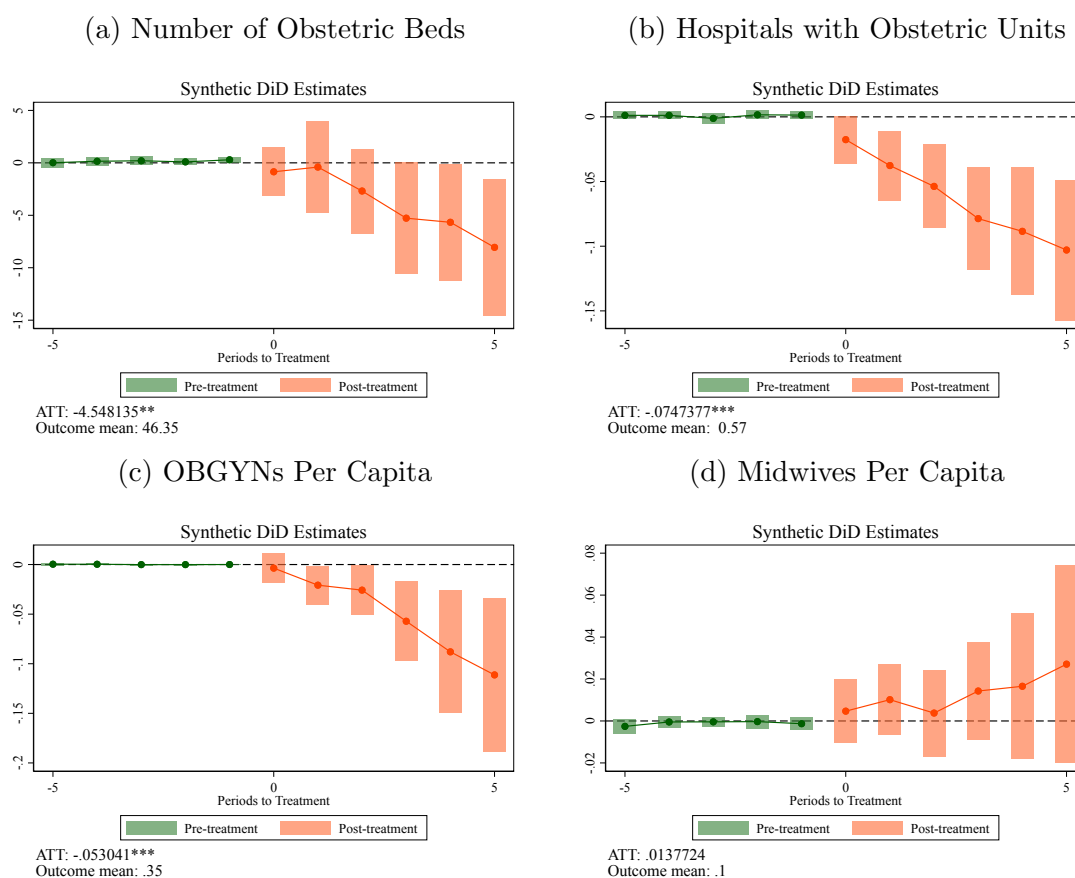
Figure 5: Event Study Estimates of the Impact of Hospital Acquisition on Births



Notes: This figure shows the event study estimates of the impact of hospital acquisition on hospital-level births.

Data source: AHA Annual Survey of Hospitals

Figure 6: Synthetic Difference-in-Difference Estimates of the Impact of Hospital Acquisition on County Level Health Care Resources

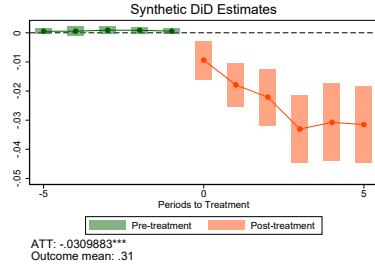


Notes: This figure shows synthetic difference-in-difference estimates of the impact of hospital acquisition on county-level obstetric resources. Subfigure (a) examines the total number of obstetric beds. Subfigure (b) examines the number of hospitals with obstetric services. Subfigure (c) examines the number of OBGYNs per capita (per 10,000). Subfigure (d) examines the number of midwives per capita (per 10,000).

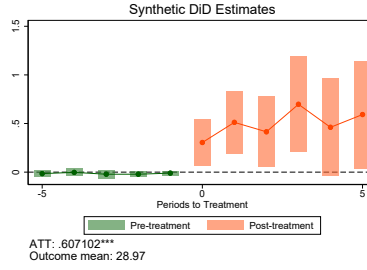
Data source: AHA Annual Survey of Hospitals, Area Health Resource Files

Figure 7: Synthetic Difference in Difference Estimates of Hospital Acquisition

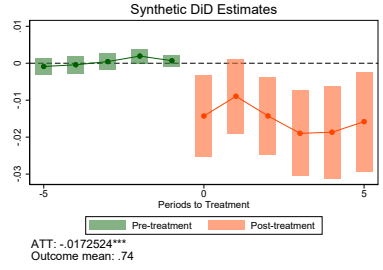
(a) Delivery in county of residence



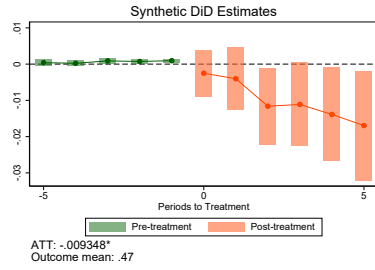
(b) Distance



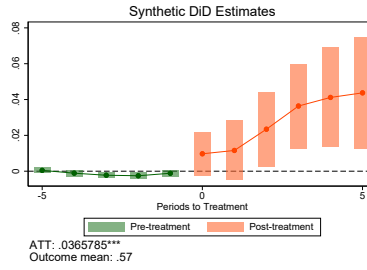
(c) First trimester prenatal care



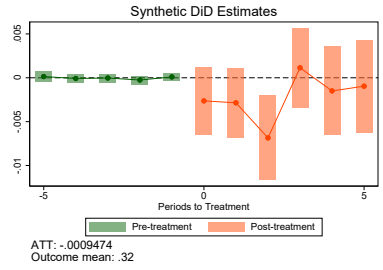
(d) Recommended care



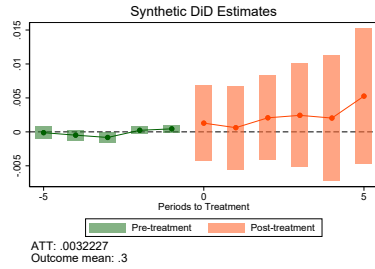
(e) Anesthesia



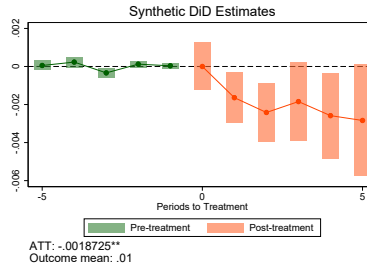
(f) Cesarean delivery



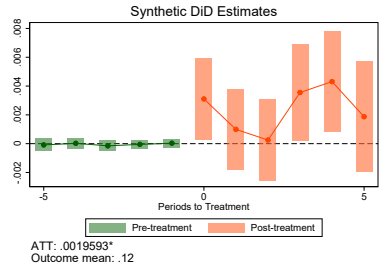
(g) Induction



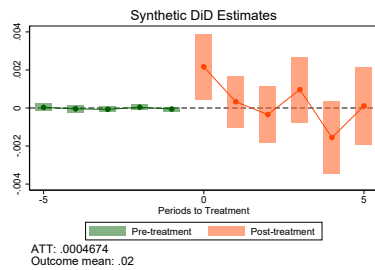
(h) Mother transferred



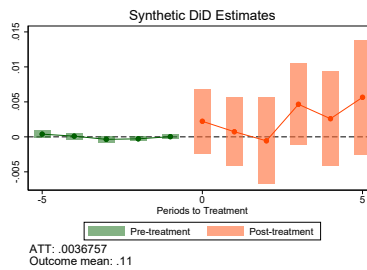
(i) Preterm birth



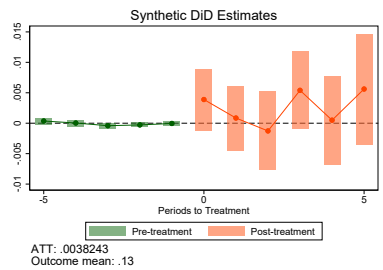
(j) Severe maternal morbidity index



(k) Chronic maternal morbidity index

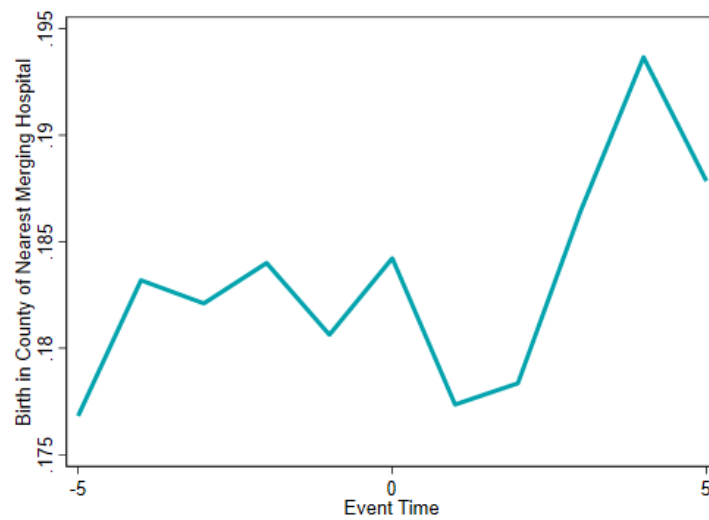


(l) Overall maternal morbidity index



Notes: This figure shows synthetic difference-in-difference estimates of the effect of a hospital being acquired in a mother's county of residence. Distance is measured in miles from the county centroid of the mother's home county to the county in which she delivers. First trimester prenatal care is defined as the first prenatal care appointment occurs within the first three months of pregnancy. Recommended care indicates whether the mother attended at least 12 prenatal visits. The severe maternal morbidity index includes maternal transfusion, perineal laceration, ruptured uterus, unplanned hysterectomy, and admission to intensive care. The chronic maternal morbidity index includes gestational hypertension, gestational diabetes, eclampsia. The overall maternal morbidity index includes both sets of conditions.
Data Source: NVSS

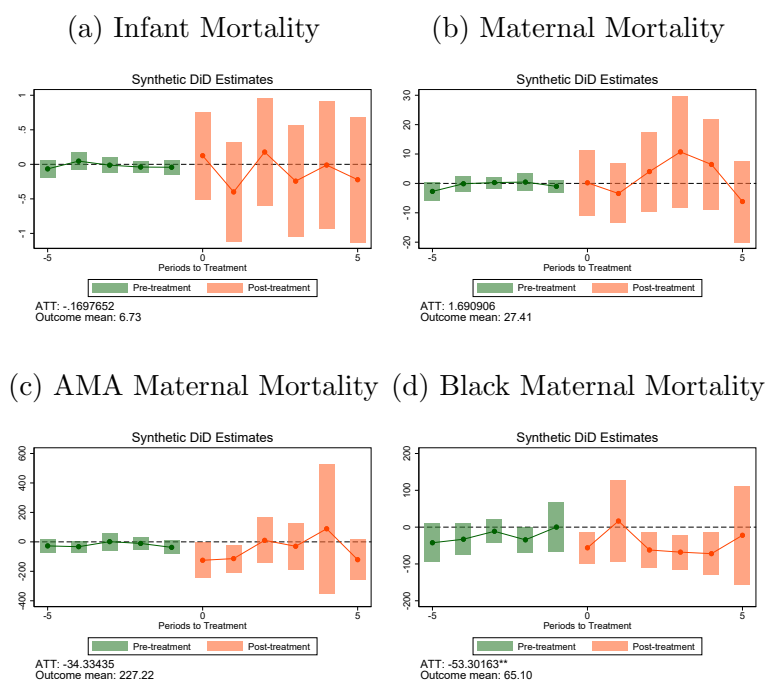
Figure 8: Births in county of nearest acquirer with obstetric unit



Notes: This figure shows the average probability that women who live in a county with an acquired hospital give birth in the county of the nearest hospital in the acquiring system with an obstetric unit.

Data source: AHA Annual Survey of Hospitals, NVSS

Figure 9: Synthetic Difference-in-Difference Estimates of the Effects of Hospital Acquisition on Infant and Maternal Mortality



Notes: This figure shows synthetic difference-in-difference estimates of the effect of a hospital being acquired in a mother's county of residence.

Data Source: NVSS

A Data Appendix

A.1 Obstetric Services

We use the AHA data to identify whether a hospital offered obstetric services. There are multiple variables that potentially identify whether a hospital offers obstetric care; we combine these variables into the following algorithm that uses information from these variables to mitigate concerns about mismeasurement that arises from mis-reporting or non-response. This algorithm combines algorithms suggested in [Interrante et al. \(2022\)](#) and [Fischer et al. \(2024\)](#). We depart from the algorithm in [Interrante et al. \(2022\)](#) in two ways: we do not use the obstetric care level variable since it originates in 2015, late in our sample window, and we additionally incorporate the variable that measures the number of staffed normal newborn bassinets, similar to [Fischer et al. \(2024\)](#).

We first classify a hospital as having obstetric care if all of the following conditions are met:

1. The hospital reports providing obstetric services
2. The hospital reports at least one dedicated obstetric bed
3. The hospital reports at least 10 births

Because the number of obstetric beds is missing for a large number of hospital years, we then classify hospitals as having obstetric services if any of the following conditions are met:

1. The hospital reports providing obstetric services and at least 10 births and is missing information on the number of dedicated obstetric beds
2. The hospital reports providing obstetric services and at least 25 births
3. The hospital reports at least 10 births and at least one normal newborn bassinet

There remain some hospitals with inconsistencies across variables in the AHA data. To resolve these inconsistencies, we cross-reference the CMS POS files. If the hospital is identified as providing obstetric care in that data source and one of our initial conditions is met in the AHA data, we classify the hospital as providing obstetric care. If the hospital reports less than 5 births and is not identified as providing obstetric care in the POS files, we classify that hospital as not having obstetric care. Additionally, for hospitals with ambiguity, if the hospital is clearly identified as providing obstetric care in years $t-1$ and $t+1$, we classify the hospital as having obstetric services in time t . Analogously, if a hospital does not provide obstetric care in $t-1$ and $t+1$ and there is remaining ambiguity about whether the hospital offers obstetric care in time t , we classify the hospital as not having obstetric services in time t . If there is remaining ambiguity, we classify the hospital as not providing obstetric services as identified in the POS files.

To address concerns that there is additional mis-reporting following mergers, we hand-check all hospitals that either change their obstetric care status or have a change in reported births of more than 50% in the 5 years following an acquisition event. We additionally hand-check hospitals that consolidate AHA reporting post-merger, as detailed in the following sub-section.

We verify our measure by comparing our results with those generated using solely the NVSS. We identify a county as having an obstetric unit if there are at least 10 hospital based births in that county in a particular year. We estimate our main event study specifications at the county-level, where the treatment is that a hospital located in that county was acquired. Appendix Figure [C11](#) shows the decline in the probability that the county has an obstetric unit following the acquisition of a hospital in the county.

A.2 Merger Identification

We primarily use the AHA data to identify merger activity through system ID changes in this data. We define a merger event as a system ID change. We then focus on acquisitions. To identify acquirers and targets for mergers where all hospitals are independent hospitals or new hospitals, we rely on the classification in the IL data. If the merger consists of existing systems, we identify the largest system as the acquiring system as measured in the AHA data.²⁰ Target hospitals are identified as the smaller party in the transaction.

There are known limitations to identifying mergers using the AHA data, as detailed in [Cooper et al. \(2019\)](#) and [Oh et al. \(2025\)](#). The primary concern is that we are misidentifying the date of the merger. To address this concern, we match system ID changes in the AHA data with the IL data. If we see the merger registered in the IL data at a later date, we use the system ID change recorded in the AHA data. If we see the merger registered in the IL data at an earlier date for a target hospital that is missing a system ID, we generate a corresponding system ID to the acquirer for the target hospital to minimize the discrepancy with IL. We hand check roughly 300 mergers that cannot be matched between the AHA and IL data.

One known issue with using the AHA data to study mergers is that when hospitals are acquired, they often report to the AHA only under the consolidated entity. To address this, we follow [Cooper et al. \(2019\)](#) and “unconsolidate” the merged entity. This is relatively rare in the rural hospital setting; we identify under which AHA identifier hospitals report using data provided by [Cooper et al. \(2019\)](#) and find only 26 instances where the rural hospital reports under an alternative identifier post-merger. For these hospitals, we hand check whether there is obstetric care provided at the acquired hospital post-merger. We treat these outcomes as missing in regressions that estimate the effects of the merger on other hospital characteristics, such as births, since it is unclear how to verify unconsolidated

²⁰There are no cases where a standalone rural hospital is larger than the combined parties in the system.

versions of these variables.

A.3 Parallel Trends

To provide additional evidence in support of the key identifying assumptions, in Appendix Figure C4 we present additional event study figures corresponding to the models estimated in Table 4. Across specifications, we find no evidence of differential trends in the pre-period taking each coefficient individually.

We then test whether our pre-period coefficients are jointly significant. Our initial test of this involves testing whether our aggregated event time pre-period coefficients are jointly equal to zero. These estimates are shown below our event study figures in Appendix Figure C4. We fail to reject the null hypothesis that these coefficients are jointly equal to zero across a variety of specifications.

We additionally can apply a more stringent test of joint significance. Recall that these coefficients are a weighted average of the group specific coefficients. We can test the joint significance of all the underlying aggregated group-time treatment effects, a very conservative test as it requires that these coefficients are jointly zero taking each cohort of treated hospitals separately. Remarkably, we fail to reject the null hypothesis here at the 1% significance level in either specifications that only use the variation from merger timing or that involve inverse probability weighting the sample (subfigures (c) and (d)).

To further address any potential concerns about violations of parallel trends, we additionally estimate a synthetic differences-in-difference model, following Arkhangelsky et al. (2021) and shown in subfigure (e). As expected, these pre-period estimates are all close to zero, and we find similar reductions in obstetrics service offerings as in our main specifications.

We additionally present event study figures examining alternative hospital characteristics including hospital margins, total payroll expenses (as a comprehensive measure of staffing), inpatient days, and general adult beds in Appendix Figure C5. We find no evidence of

differential pre-trends in any of these categories.

While balance on observable characteristics at the time of acquisition is not necessary for parallel trends to hold, we test for selection on observable characteristics into acquisition in several ways. In Table 2, we report differences between treated and untreated hospitals at baseline, but that may fail to account for changes over time in the lead up to acquisition. In Appendix Figure C1, we present balancing regressions that show the differences in hospital and county characteristics between acquired and non-acquired hospitals at the point of acquisition once we account for time and state fixed effects. We find no differences between acquired and never-acquired rural hospitals in terms of financial characteristics but do find higher patient utilization and more service offerings at acquired hospitals. We find that counties where a hospital was acquired have larger populations and more OBGYNs per capita on average, but otherwise look similar on socio-demographic characteristics.

We additionally test whether we can “predict” acquisition: we regress acquisition on state and time fixed effects and a vector of hospital characteristics. We show estimated coefficients from this model in Appendix Table B1. This model has low predictive power. Government hospitals, critical access hospitals, and hospitals already in systems are less likely to be acquired. We additionally find a small negative relationship between margins and total admissions, though the effect sizes are small and imprecisely estimated.

Finally, we also estimate our event study specification at the county level using characteristics we think are unlikely to have been directly affected by the hospital merger to test whether the counties in which acquired hospitals are located were trending differently than non-acquired hospitals. As we detail in Section 4, we find little evidence of large changes in demographic characteristics.

B Additional Tables

Table B1: Linear Prediction Model of Acquisition

	(1) Acquired in merger
L.Margin (Income - Cost) / Income	-0.0066* (0.0040)
L.Non Profit	0.0149*** (0.0029)
L.For Profit	0.0354*** (0.0046)
L.General Adult Beds	0.0110 (0.0075)
L.In System	-0.0143*** (0.0025)
L.Critical Access Hospital	-0.0059* (0.0031)
L.Payroll Expenses	-0.0000 (0.0009)
L.Total Births	-0.0007 (0.0077)
L.Total Admissions	-0.0024* (0.0014)
L.ER	0.0066 (0.0099)
L.OB Services	0.0018 (0.0031)
Constant	0.0034 (0.0107)
N	18,669
R2	0.0167

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports results from a regression of acquisition on hospital characteristics, year, and state fixed effects. Hospital beds, births, and admissions reported in 1000s. Payroll expenses are in millions.

Table B2: Difference-in-Difference Estimates of the Effects of Hospital Acquisition on Obstetric Units

	(1) Independent in 2007	(2) Acquirers in Control	(3) No Acquirers	(4) Include New Hospitals/Systems
ATT	-0.0336** (0.0159)	-0.0500*** (0.0137)	-0.0430 (0.0302)	-0.0418*** (0.0127)
N	13687	24824	12718	24750
Outcome Mean	.54	.58	.52	.58

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table presents our estimated aggregated average treatment effects for the effect of hospital acquisition on obstetric units. Column 1 reports our estimates from a restricted sample that consists of only hospitals that were independent as of 2007. Column 2 reports estimates from a sample that allows hospitals in both the treatment and control group to be acquirers. Column 3 restricts the sample to no acquirers in either treatment or control group. Column 4 includes mergers where we cannot identify a target hospital.

Data source: AHA Annual Survey of Hospitals

Table B3: Number of Acquired Hospitals By Year

	(1)
2007	23
2008	24
2009	40
2010	36
2011	46
2012	54
2013	43
2014	54
2015	34
2016	40
2017	43
2018	42
2019	20
Total	499
Observations	499

Notes: This table shows the number of treated hospitals by year. Treated is defined as being acquired for the first time in our sample period. Some hospitals will be acquired multiple times.

Table B4: Average Treatment Effect By Cohort

	(1)
GAverage	-0.0453*** (0.0112)
G2008	-0.0593 (0.0553)
G2009	0.00216 (0.0320)
G2010	-0.194*** (0.0571)
G2011	-0.0498 (0.0433)
G2012	0.0204 (0.0161)
G2013	-0.0358 (0.0300)
G2014	-0.0518 (0.0412)
G2015	-0.0818* (0.0463)
G2016	-0.0407 (0.0311)
G2017	-0.00590 (0.0165)
G2018	-0.0429 (0.0355)
G2019	-0.0517 (0.0541)
Observations	16615

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows our estimated average treatment effects by cohorts of acquisition year.

Table B5: Effects of Acquisition on Other Service Line Offerings

	(1) ER	(2) Psych	(3) Substance	(4) NICU	(5) PICU
ATT	-0.00943* (0.00568)	-0.0195 (0.0140)	-0.0183* (0.0100)	-0.0100 (0.0102)	-0.00367 (0.00816)
N	23014	23013	23017	23019	23019
Outcome Mean	.98	.15	.04	.04	.03

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows the estimated effect of hospital acquisition on other service line offerings.

Table B6: Hospital Referral Region Hospital Resources

	(1)	(2)	(3)	(4)
	HRR OB Units	HRR OB Beds	Unacquired Unit	Unacquired Beds
ATT	-0.410*** (0.113)	-57.71** (24.97)	-0.00983 (0.00857)	-5.705 (3.615)
N	3705	3705	32117	31346
Outcome Mean	8.99	1732.11	.68	128.16

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table estimates regressions examining the effects of hospital acquisition on regional health care resources, measured at the HRR level.

Table B7: Effects of Acquisition on County Level Resources

	(1) Main CS	(2) Only Hosp	(3) Not Yet	(4) No Acquirers	(5) No Acq. in Control	(6) No Micro	(7) Synthetic
<i>OB Beds</i>							
ATT	-4.516** (1.948)	-4.044** (1.980)	-4.447** (1.950)	-2.195 (2.761)	-5.183*** (1.933)	-2.791 (2.153)	-4.548** (2.108)
N	26754	21463	26754	15769	20241	18356	26754
Outcome Mean	46.35	57.77	46.35	25.79	40.13	25.16	46.35
<i>OB Hospitals</i>							
ATT	-0.0767*** (0.0165)	-0.0661*** (0.0167)	-0.0751*** (0.0165)	-0.0200 (0.0274)	-0.0840*** (0.0165)	-0.0762*** (0.0213)	-0.0747*** (0.0172)
N	26754	21463	26754	15769	20241	18356	26754
Outcome Mean	.57	.72	.57	.37	.5	.41	.57
<i>OBGYNs Per Cap</i>							
ATT	-0.0547*** (0.0145)	-0.0528*** (0.0148)	-0.0540*** (0.0144)	-0.0821*** (0.0263)	-0.0525*** (0.0147)	-0.0656*** (0.0202)	-0.0530*** (0.0161)
N	17595	13932	17595	10872	13086	12114	17595
Outcome Mean	.35	.43	.35	.26	.31	.21	.35
<i>Midwives Per Cap</i>							
ATT	0.0138 (0.0115)	0.00968 (0.0117)	0.0133 (0.0115)	0.0218 (0.0149)	0.0169 (0.0117)	0.0128 (0.0130)	0.0138 (0.0119)
N	17565	13931	17565	10842	13056	12089	17325
Outcome Mean	.1	.12	.1	.08	.09	.07	.1

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows estimates from different specification choices on county-level health care resources. Column (1) shows our main Callaway Sant'Anna estimates. Column (2) restricts to a sample that contains only counties with a hospital in at least one year. Column (3) includes not yet treated hospitals in estimating time trends in the control group. Column (4) drops all counties that have a hospital that is part of an acquiring system at some point in the sample window. Column (5) drops counties with an acquiring system from the control group. Column (6) drops counties in micropolitan areas. Column (7) reports synthetic difference in difference estimates.

Table B8: Effects of Acquisition on Delivery Location, Prenatal Care, and Interventions

	(1) Main CS	(2) Only Hosp	(3) Not Yet	(4) No Acquirers	(5) No Acq. in Control	(6) No Micro	(7) Synthetic
<i>Delivery in county of residence</i>							
ATT	-0.0312*** (0.00542)	-0.0245*** (0.00552)	-0.0304*** (0.00541)	-0.0309*** (0.0119)	-0.0360*** (0.00546)	-0.0273*** (0.00775)	-0.0310*** (0.00493)
N	26611	21411	26611	15639	20111	18213	26611
Outcome Mean	.31	.37	.31	.22	.26	.2	.31
<i>Distance</i>							
ATT	0.585*** (0.222)	0.477** (0.227)	0.561** (0.222)	0.587 (0.436)	0.622*** (0.226)	0.417 (0.330)	0.607*** (0.210)
N	26611	21411	26611	15639	20111	18213	26611
Outcome Mean	28.97	26	28.97	33.64	30.94	34.19	28.97
<i>First trimester prenatal</i>							
ATT	-0.0122** (0.00617)	-0.0131** (0.00621)	-0.0121* (0.00618)	0.00197 (0.00766)	-0.0116* (0.00623)	-0.00770 (0.00845)	-0.0173*** (0.00584)
N	19630	16003	19630	11388	14677	13052	19630
Outcome Mean	.74	.74	.74	.74	.74	.74	.74
<i>Recommended prenatal visits</i>							
ATT	-0.00247 (0.00506)	-0.00188 (0.00513)	-0.00194 (0.00506)	0.00222 (0.0115)	-0.00301 (0.00516)	-0.00227 (0.00640)	-0.00935* (0.00479)
N	26611	21411	26611	15639	20111	18213	26611
Outcome Mean	.47	.48	.47	.46	.47	.46	.47
<i>Cesarean</i>							
ATT	-0.00282 (0.00216)	-0.00335 (0.00215)	-0.00290 (0.00216)	-0.00389 (0.00497)	-0.00233 (0.00223)	-0.00363 (0.00310)	-0.000947 (0.00201)
N	26611	21411	26611	15639	20111	18213	26611
Outcome Mean	.32	.32	.32	.32	.32	.32	.32
<i>Induction</i>							
ATT	-0.000360 (0.00342)	-0.00172 (0.00346)	-0.000843 (0.00342)	0.00447 (0.00904)	0.000116 (0.00351)	-0.0000147 (0.00475)	0.00322 (0.00376)
N	26611	21411	26611	15639	20111	18213	26611
Outcome Mean	.3	.3	.3	.3	.3	.3	.3
<i>Anesthesia</i>							
ATT	0.0302** (0.0121)	0.0326*** (0.0122)	0.0289** (0.0121)	-0.00659 (0.0166)	0.0291** (0.0123)	0.0359** (0.0153)	0.0366*** (0.0120)
N	26611	21411	26611	15639	20111	18213	26611
Outcome Mean	.57	.57	.57	.57	.57	.57	.57
<i>Mother transferred</i>							
ATT	-0.00226*** (0.000874)	-0.00217** (0.000920)	-0.00223*** (0.000863)	-0.00275** (0.00124)	-0.00286*** (0.00102)	-0.00217* (0.00118)	-0.00187** (0.000865)
N	21791	17391	21791	13189	16291	14916	21791
Outcome Mean	.01	.01	.01	.02	.01	.02	.01
<i>Preterm</i>							
ATT	0.000500 (0.00139)	0.000487 (0.00139)	0.000434 (0.00139)	0.00287 (0.00341)	0.0000116 (0.00143)	0.00152 (0.00209)	0.00196* (0.00115)
N	26611	21411	26611	15639	20111	18213	26611
Outcome Mean	.12	.12	.12	.12	.12	.12	.12

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows estimates from different specification choices on patient delivery locations, prenatal care, and interventions. Column (1) shows our main Callaway Sant'Anna estimates. Column (2) restricts to a sample that contains only counties with a hospital in at least one year. Column (3) includes not yet treated hospitals in estimating time trends in the control group. Column (4) drops all counties that have a hospital that is part of an acquiring system at some point in the sample window. Column (5) drops counties with an acquiring system from the control group. Column (6) drops counties in micropolitan areas. Column (7) reports synthetic difference in difference estimates.

Table B9: Effects of Acquisition on Maternal Morbidity

	(1) Main CS	(2) Only Hosp	(3) Not Yet	(4) No Acquirers	(5) No Acq. in Control	(6) No Micro	(7) Synthetic
<i>Any SMM</i>							
ATT	-0.000428 (0.000889)	-0.000393 (0.000893)	-0.000442 (0.000894)	-0.00141 (0.00145)	-0.000439 (0.000904)	-0.000101 (0.00129)	0.000319 (0.000640)
N	21791	17391	21791	13189	16291	14916	21791
Outcome Mean	.01	.01	.01	.01	.01	.01	.01
<i>SMM sum</i>							
ATT	-0.000356 (0.000929)	-0.000366 (0.000934)	-0.000368 (0.000934)	-0.00156 (0.00162)	-0.000373 (0.000945)	0.0000255 (0.00136)	0.000467 (0.000662)
N	21791	17391	21791	13189	16291	14916	21791
Outcome Mean	.02	.02	.02	.02	.02	.02	.02
<i>Any MM</i>							
ATT	0.0000486 (0.00278)	0.000169 (0.00280)	-0.0000615 (0.00278)	-0.00120 (0.00475)	-0.000135 (0.00284)	0.00267 (0.00406)	0.00341 (0.00264)
N	21791	17391	21791	13189	16291	14916	21791
Outcome Mean	.11	.12	.11	.11	.11	.11	.11
<i>MM sum</i>							
ATT	0.000211 (0.00302)	0.000268 (0.00305)	0.000107 (0.00302)	-0.00176 (0.00522)	-0.0000314 (0.00310)	0.00290 (0.00441)	0.00382 (0.00285)
N	21791	17391	21791	13189	16291	14916	21791
Outcome Mean	.13	.13	.13	.12	.12	.13	.13
<i>Any chronic condition</i>							
ATT	0.000392 (0.00246)	0.000477 (0.00249)	0.000309 (0.00246)	-0.000120 (0.00452)	0.000189 (0.00253)	0.00254 (0.00358)	0.00346 (0.00244)
N	21791	17391	21791	13189	16291	14916	21791
Outcome Mean	.1	.1	.1	.1	.1	.1	.1
<i>Chronic sum</i>							
ATT	0.000566 (0.00262)	0.000633 (0.00265)	0.000475 (0.00262)	-0.000202 (0.00484)	0.000341 (0.00270)	0.00288 (0.00379)	0.00368 (0.00262)
N	21791	17391	21791	13189	16291	14916	21791
Outcome Mean	.11	.11	.11	.11	.11	.11	.11

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows estimates from different specification choices on maternal morbidity. Column (1) shows our main Callaway Sant'Anna estimates. Column (2) restricts to a sample that contains only counties with a hospital in at least one year. Column (3) includes not yet treated hospitals in estimating time trends in the control group. Column (4) drops all counties that have a hospital that is part of an acquiring system at some point in the sample window. Column (5) drops counties with an acquiring system from the control group. Column (6) drops counties in micropolitan areas. Column (7) reports synthetic difference in difference estimates.

Table B10: Difference-in-Difference Estimates of the Effects of Hospital Acquisition on Mortality

	(1) Main CS	(2) Only Hosp	(3) Not Yet	(4) No Acquirers	(5) No Acq. in Control	(6) No Micro	(7) Synthetic
<i>Infant mortality rate</i>							
ATT	-0.321 (0.337)	-0.302 (0.322)	-0.320 (0.337)	-0.0800 (0.998)	-0.276 (0.360)	-0.589 (0.507)	-0.170 (0.305)
N	26717	21461	26717	15735	20207	18303	26611
Outcome Mean	6.77	6.71	6.77	6.85	6.85	6.85	6.73
<i>Maternal mortality rate</i>							
ATT	-15.46* (8.407)	-11.91 (8.328)	-15.54* (8.455)	-9.264 (17.48)	-17.25** (8.647)	-19.27 (13.28)	1.691 (4.257)
N	26709	21461	26709	15727	20199	18296	26572
Outcome Mean	28.11	27.63	28.11	30.57	29.15	29.84	27.41
<i>Black maternal mortality rate</i>							
ATT	-168.4 (130.0)	-173.0 (130.6)	-174.2 (135.2)	-64.16 (50.37)	-196.1 (132.4)	-278.6 (250.6)	-53.30** (26.73)
N	18406	15996	18540	9678	13279	10908	12194
Outcome Mean	66.77	72.3	66.29	85.54	79.39	73.15	65.1
<i>AMA maternal mortality rate</i>							
ATT	-178.6 (142.0)	-154.8 (142.3)	-178.7 (142.5)	-80.03 (228.0)	-183.3 (143.5)	-180.4 (233.2)	-34.33 (47.71)
N	25913	21167	25933	15036	19437	17555	23530
Outcome Mean	235.12	233.65	234.94	272.27	249.85	261.35	227.22

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table presents our estimated aggregated average treatment effects for the effect of hospital acquisition on mortality. Column (1) shows our main Callaway Sant'Anna estimates. Column (2) restricts to a sample that contains only counties with a hospital in at least one year. Column (3) includes not yet treated hospitals in estimating time trends in the control group. Column (4) drops all counties that have a hospital that is part of an acquiring system at some point in the sample window. Column (5) drops counties with an acquiring system from the control group. Column (6) drops counties in micropolitan areas. Column (7) reports synthetic difference in difference estimates. Data source: NVSS

Table B11: Effects of Acquisition on Maternal Characteristics

	(1) Main CS	(2) Only Hosp	(3) Not Yet	(4) No Acquirers	(5) No Acq. in Control	(6) No Micro	(7) Synthetic
<i>Medicaid</i>							
ATT	0.00984 (0.0103)	0.00788 (0.0103)	0.00934 (0.0102)	-0.00467 (0.0108)	0.0115 (0.0103)	-0.00459 (0.0129)	0.0128 (0.0107)
N	18029	14608	18029	10780	13321	12111	18029
Outcome Mean	.46	.46	.46	.46	.46	.46	.47
<i>Maternal age</i>							
ATT	0.00533 (0.0259)	-0.00641 (0.0258)	0.00508 (0.0259)	-0.0250 (0.0484)	0.0000394 (0.0267)	0.0176 (0.0392)	-0.00639 (0.0211)
N	26611	21411	26611	15639	20111	18213	26611
Outcome Mean	26.66	26.63	26.66	26.65	26.63	26.65	26.66
<i>White</i>							
ATT	-0.00504*** (0.00174)	-0.00435** (0.00174)	-0.00494*** (0.00174)	0.000461 (0.00380)	-0.00586*** (0.00177)	-0.00654** (0.00265)	-0.00637*** (0.00165)
N	26611	21411	26611	15639	20111	18213	26611
Outcome Mean	.86	.86	.86	.85	.85	.86	.86
<i>College+</i>							
ATT	0.00118 (0.00364)	0.000635 (0.00367)	0.000941 (0.00363)	-0.00126 (0.00487)	0.00266 (0.00369)	0.00589 (0.00489)	0.00382 (0.00376)
N	26611	21411	26611	15639	20111	18213	26611
Outcome Mean	.17	.17	.17	.17	.17	.17	.17
<i>Less than high school</i>							
ATT	0.0108** (0.00457)	0.0111** (0.00463)	0.0102** (0.00457)	0.00167 (0.00648)	0.0114** (0.00466)	0.0118* (0.00608)	0.0152*** (0.00432)
N	26611	21411	26611	15639	20111	18213	26611
Outcome Mean	.14	.14	.14	.14	.14	.14	.14

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows estimates from different specification choices on maternal characteristics. Column (1) shows our main Callaway Sant'Anna estimates. Column (2) restricts to a sample that contains only counties with a hospital in at least one year. Column (3) includes not yet treated hospitals in estimating time trends in the control group. Column (4) drops all counties that have a hospital that is part of an acquiring system at some point in the sample window. Column (5) drops counties with an acquiring system from the control group. Column (6) drops counties in micropolitan areas. Column (7) reports synthetic difference in difference estimates.

Table B12: Effects of Acquisition on County Demographics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Main CS	Only Hosp	Not Yet	No Acquirers	No Acq. in Control	No Micro	Synthetic
<i>Log(Population)</i>							
ATT	-0.000253 (0.00202)	-0.00140 (0.00204)	-0.0000991 (0.00200)	0.00234 (0.00349)	0.000743 (0.00216)	-0.00256 (0.00286)	-0.000444 (0.00191)
N	21912	17435	21912	13299	16401	15037	21912
Outcome Mean	9.64	9.89	9.64	9.32	9.51	9.31	9.64
<i>Median Income</i>							
ATT	-136.6 (153.1)	-146.4 (154.6)	-115.3 (152.6)	-4.508 (243.6)	-144.8 (161.6)	71.13 (228.0)	-189.9 (156.0)
N	21911	17434	21911	13298	16400	15037	21901
Outcome Mean	43520.33	43750.32	43520.33	43246.01	43282.28	42884.97	43522.52
<i>% Pop 18-34</i>							
ATT	-0.00135** (0.000572)	-0.00174*** (0.000551)	-0.00130** (0.000566)	-0.00159* (0.000943)	-0.00165*** (0.000633)	-0.00189** (0.000862)	-0.00136** (0.000627)
N	21912	17435	21912	13299	16401	15037	21912
Outcome Mean	.2	.2	.2	.19	.2	.19	.2
<i>% Pop White</i>							
ATT	-0.000910 (0.00108)	-0.000438 (0.00110)	-0.000904 (0.00107)	-0.00226 (0.00148)	-0.00154 (0.00120)	-0.00109 (0.00137)	-0.00164 (0.00118)
N	21912	17435	21912	13299	16401	15037	21912
Outcome Mean	.85	.85	.85	.84	.84	.85	.85
<i>% Uninsured</i>							
ATT	-0.00193 (0.00135)	-0.00281** (0.00135)	-0.00187 (0.00134)	-0.000952 (0.00213)	-0.00157 (0.00141)	-0.00345** (0.00176)	-0.00191 (0.00144)
N	12712	9863	12712	8351	9205	8911	12712
Outcome Mean	.13	.13	.13	.14	.13	.13	.13

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows estimates from different specification choices on county demographics. Column (1) shows our main Callaway Sant'Anna estimates. Column (2) restricts to a sample that contains only counties with a hospital in at least one year. Column (3) includes not yet treated hospitals in estimating time trends in the control group. Column (4) drops all counties that have a hospital that is part of an acquiring system at some point in the sample window. Column (5) drops counties with an acquiring system from the control group. Column (6) drops counties in micropolitan areas. Column (7) reports synthetic difference in difference estimates.

Table B13: Only Counties Without Closures

	(1)	(2)	(3)	(4)
	OB Beds	OB Hospitals	OBGYNS Per Cap	Midwives Per Cap
ATT	-4.439** (1.964)	-0.0792*** (0.0175)	-0.0595*** (0.0157)	0.0120 (0.0129)
N	25129	25129	16551	16521
Outcome Mean	44.93	.57	.35	.1

Standard errors in parentheses

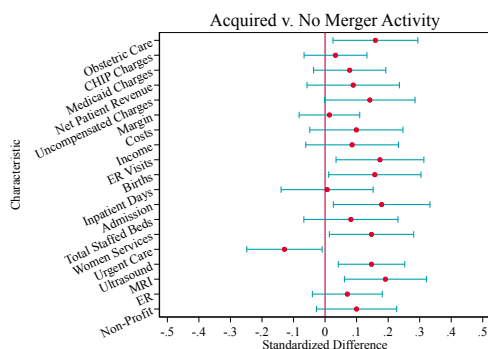
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows estimates of the effects of acquisitions conditioning on no hospitals closing in the county.

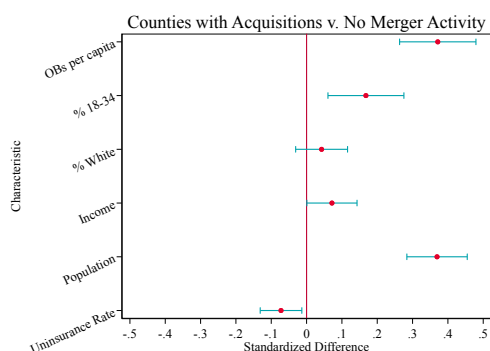
C Additional Figures

Figure C1: Balancing Regression Estimates

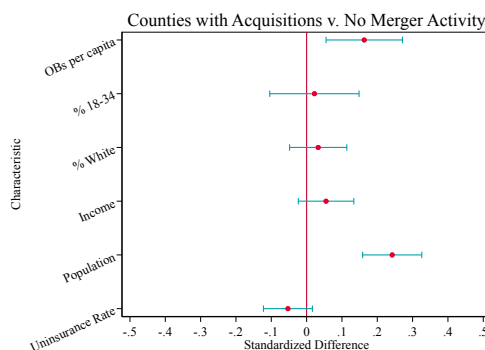
(a) Hospital Characteristics



(b) County Characteristics



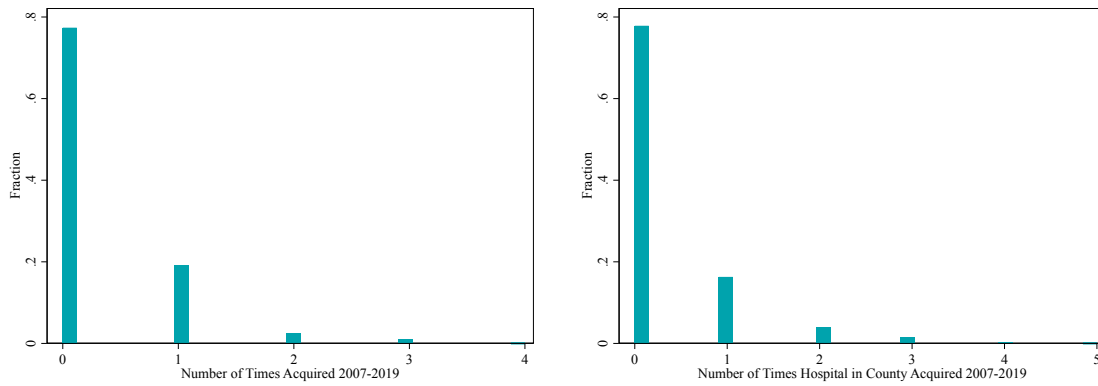
(c) County Characteristics - Just Counties with Hospitals



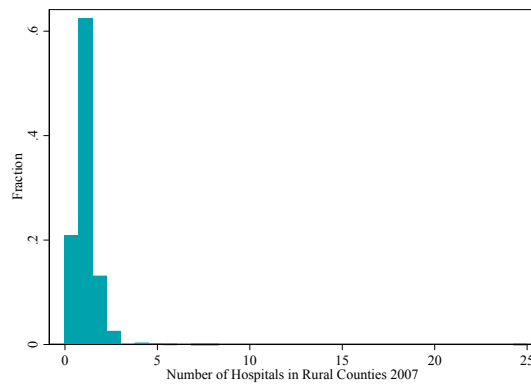
Notes: This figure shows the estimated differences in characteristics between acquired and non-acquired hospitals (subfigure a) and between counties with acquired hospitals and without acquired hospitals (subfigure b) after time and state fixed effects have been residualized. Data source: AHA Annual Survey of Hospitals, American Community Survey

Figure C2: Repeat Acquisitions

(a) Distribution of Hospital Acquisitions (b) Distribution of Acquisitions in Counties



(c) Distribution of Hospitals in Counties



Notes: This figure shows the number of times each hospital is acquired (subfigure a), that a county experiences the acquisition of a hospital located in it (subfigure b), and the number of hospitals in each county in 2007 (subfigure c).

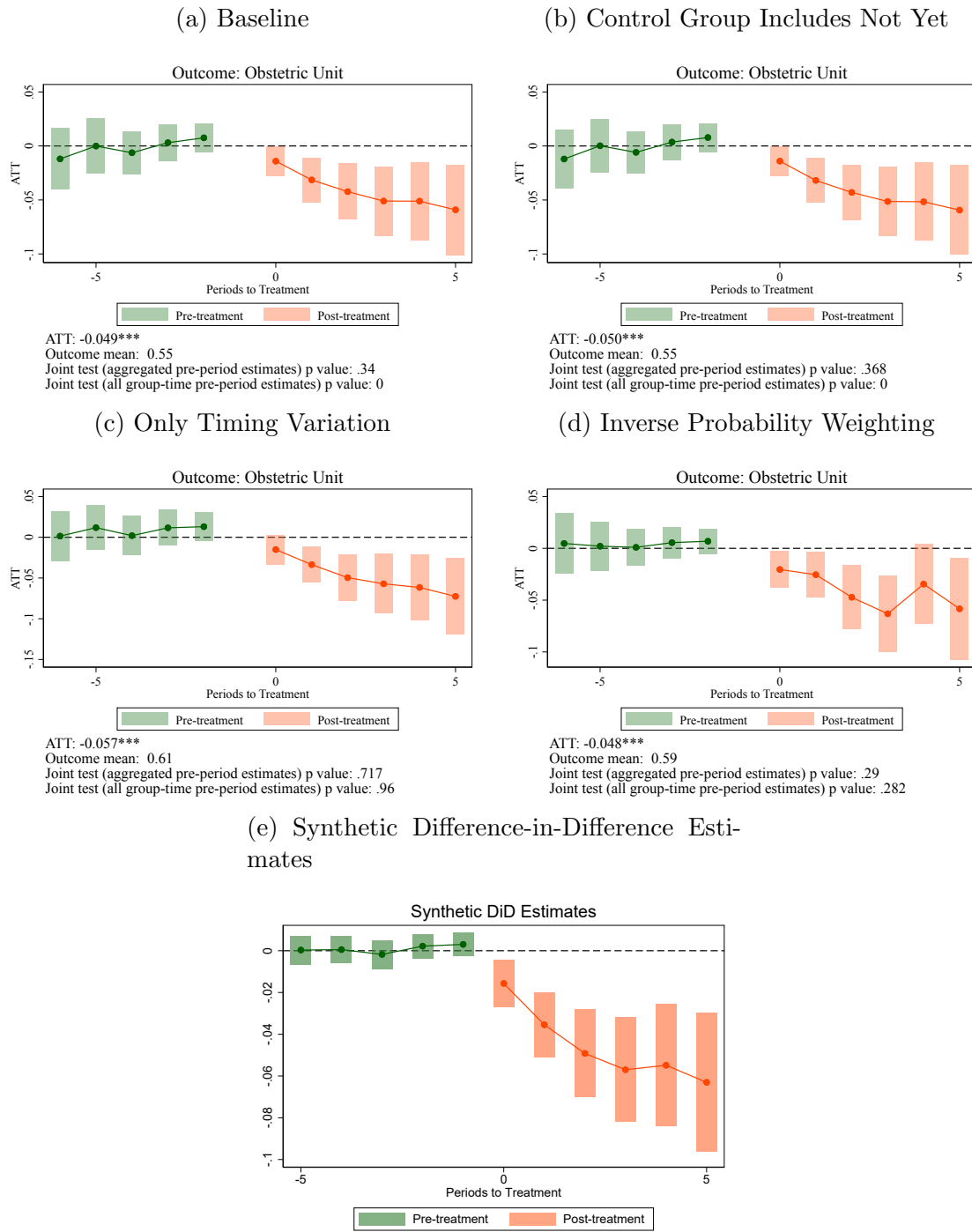
Figure C3: Estimates For Different Treatment Groups



Notes: This figure shows the event study estimates of the impact of hospital acquisition in the county of residence on whether the hospital offers obstetrics services.

Data source: AHA Annual Survey of Hospitals

Figure C4: Event Study Estimates of the Impact of Hospital Acquisition on Obstetric Units



Notes: This figure shows the event study estimates of the impact of hospital acquisition on whether the hospital offers obstetrics services.
Data source: AHA Annual Survey of Hospitals

Figure C5: Event Study Estimates of the Impact of Hospital Acquisition on Hospital Characteristics

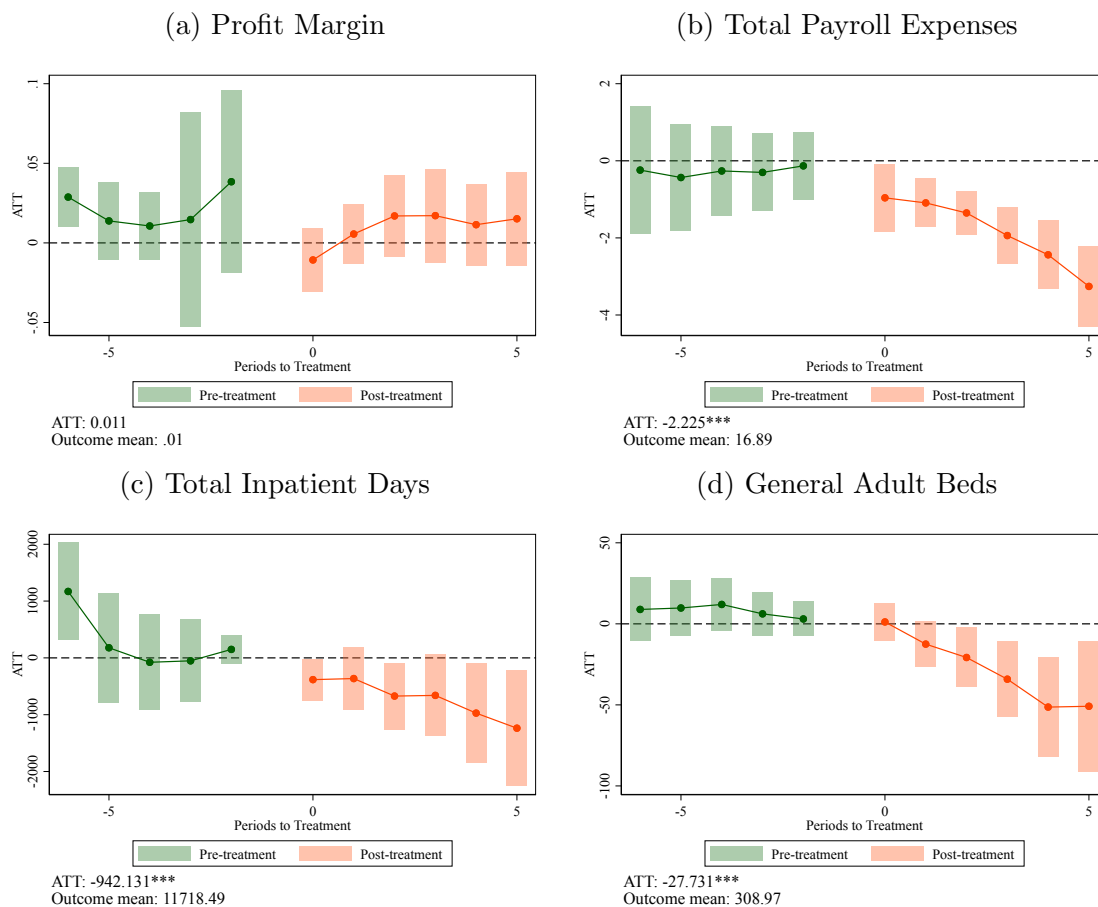


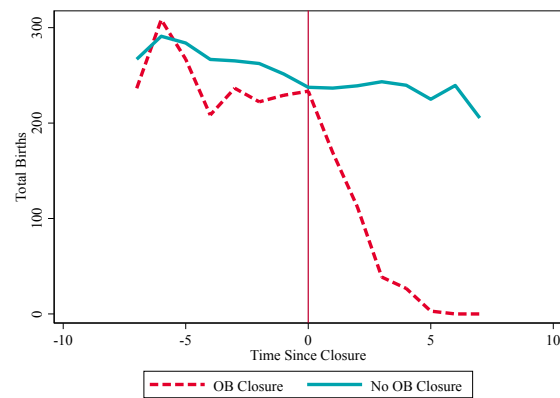
Figure C6: Event Study Estimates of the Impact of Hospital Acquisition on County Level Health Care Resources



Notes: This figure shows the event study estimates of the impact of hospital acquisition on county-level obstetric resources. Subfigure (a) examines the total number of obstetric beds. Subfigure (b) examines the number of hospitals with obstetric services. Subfigure (c) examines the number of OBGYNs per capita (per 10,000). Subfigure (d) examines the number of midwives per capita (per 10,000).

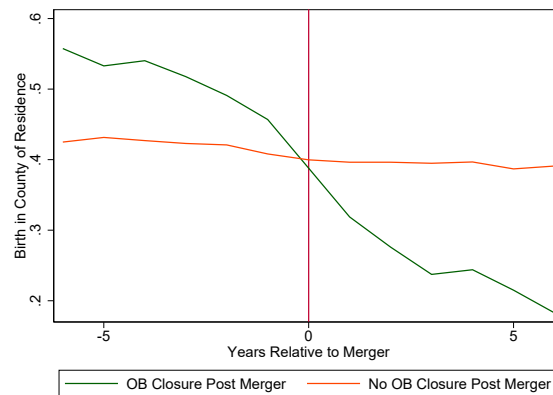
Data source: AHA Annual Survey of Hospitals, Area Health Resource Files

Figure C7: Births at Acquired Hospitals



Notes: This figure shows the time series of births at hospitals that were acquired split by whether an obstetric unit closure followed the acquisition or not.

Figure C8: Births in county of residence by closure status

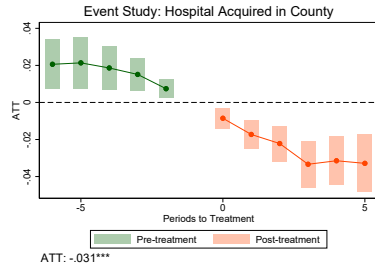


Notes: This figure shows the average probability that women who live in a county with an acquired hospital give birth in their county of residence, split by whether there is a closure of the obstetric unit in the 7 years following the acquisition.

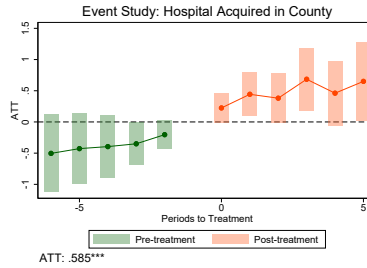
Data source: NVSS

Figure C9: Event Study Estimates of Hospital Acquisition

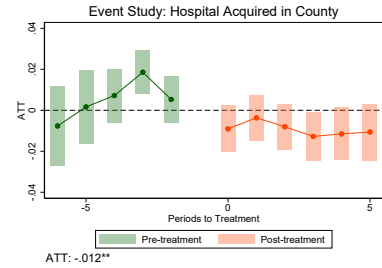
(a) Delivery in County of Residence



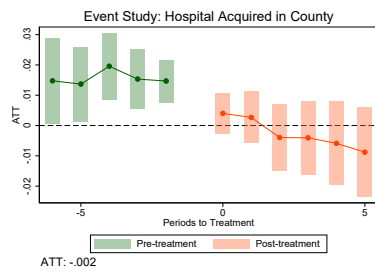
(b) Distance



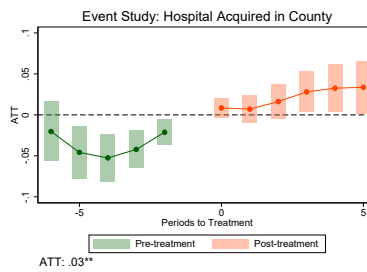
(c) First Trimester Prenatal Care



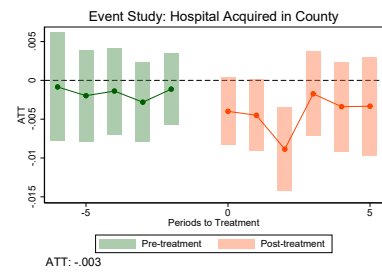
(d) Recommended Care



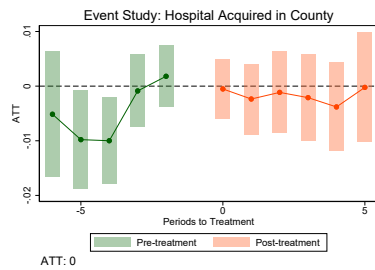
(e) Anesthesia



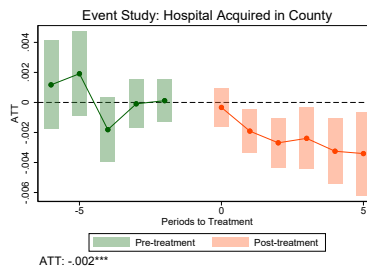
(f) Cesarean delivery



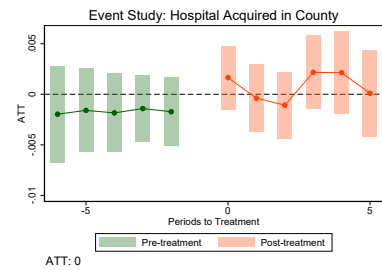
(g) Induction



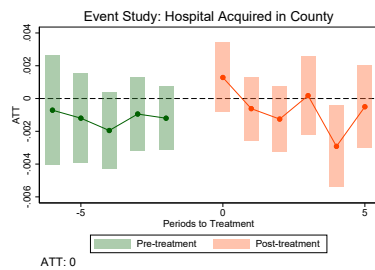
(h) Mother Transferred



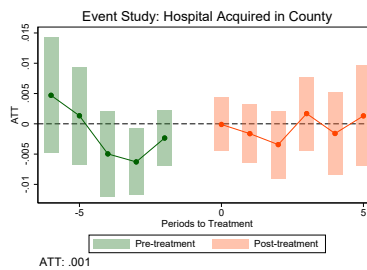
(i) Preterm birth



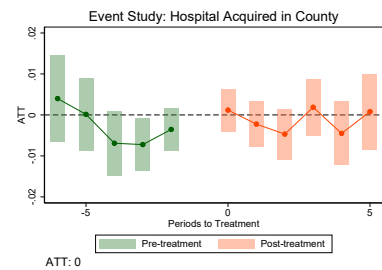
(j) Severe Maternal Morbidity Index



(k) Chronic Maternal Morbidity Index



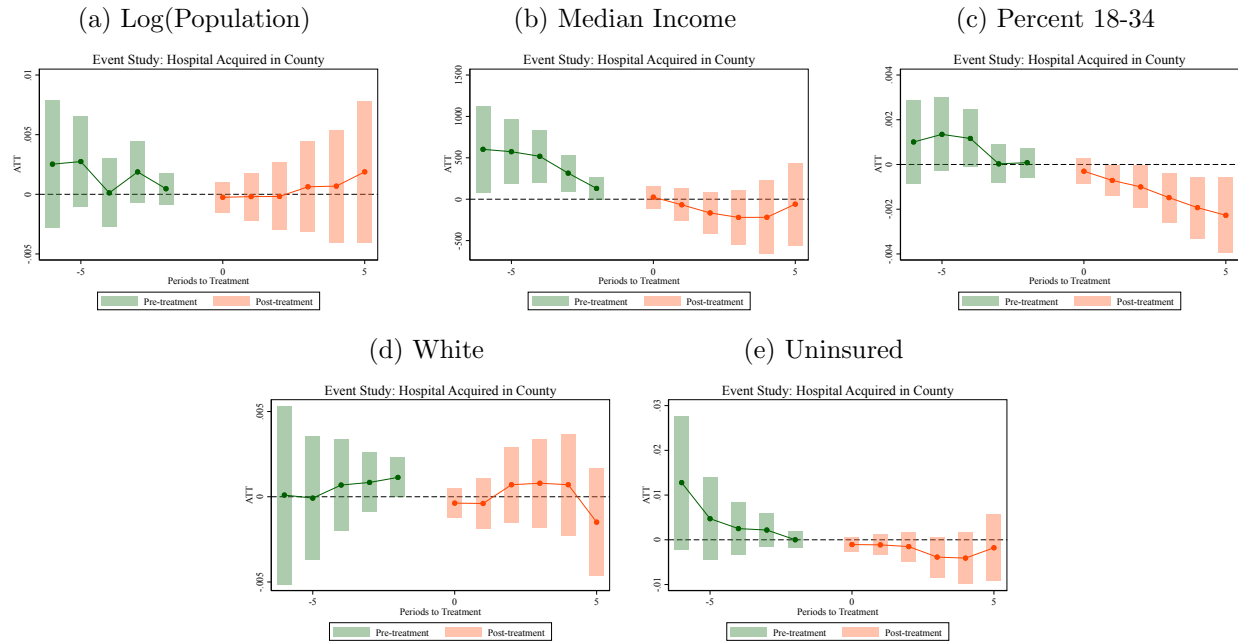
(l) Overall Maternal Morbidity Index



Notes: This figure shows Callaway Sant'Anna estimates of the effect of a hospital being acquired in a mother's county of residence.

Data Source: NVSS

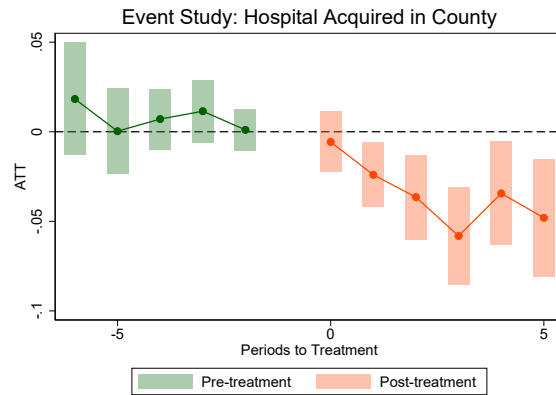
Figure C10: Event Study Estimates of Effects on County Characteristics



Notes: This figure shows Callaway Sant'Anna estimates of the effect of a hospital being acquired in a mother's county of residence.

Data Source: ACS

Figure C11: Alternative Definition of Obstetric Units Using NVSS Data



Notes: This figure shows an event study plot of the probability that a county has an obstetric unit, defined as 10 or more births in that county using the NVSS data.
Data source: NVSS